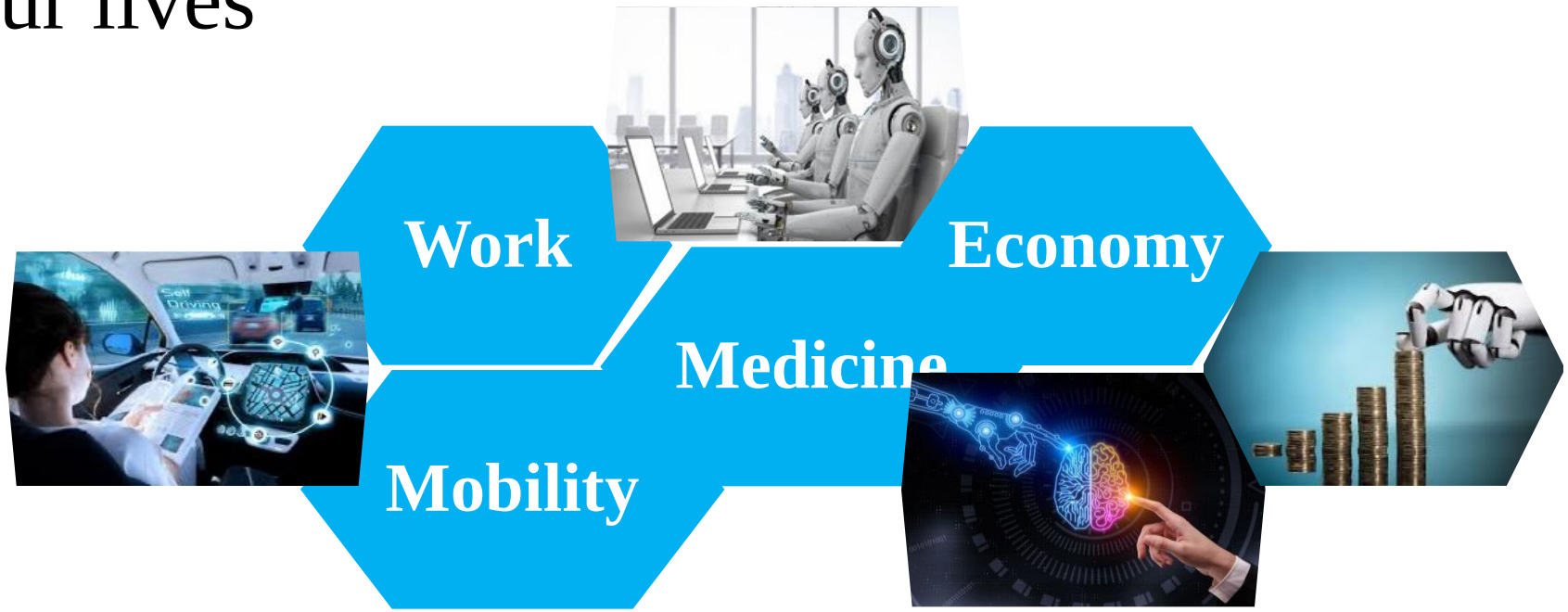
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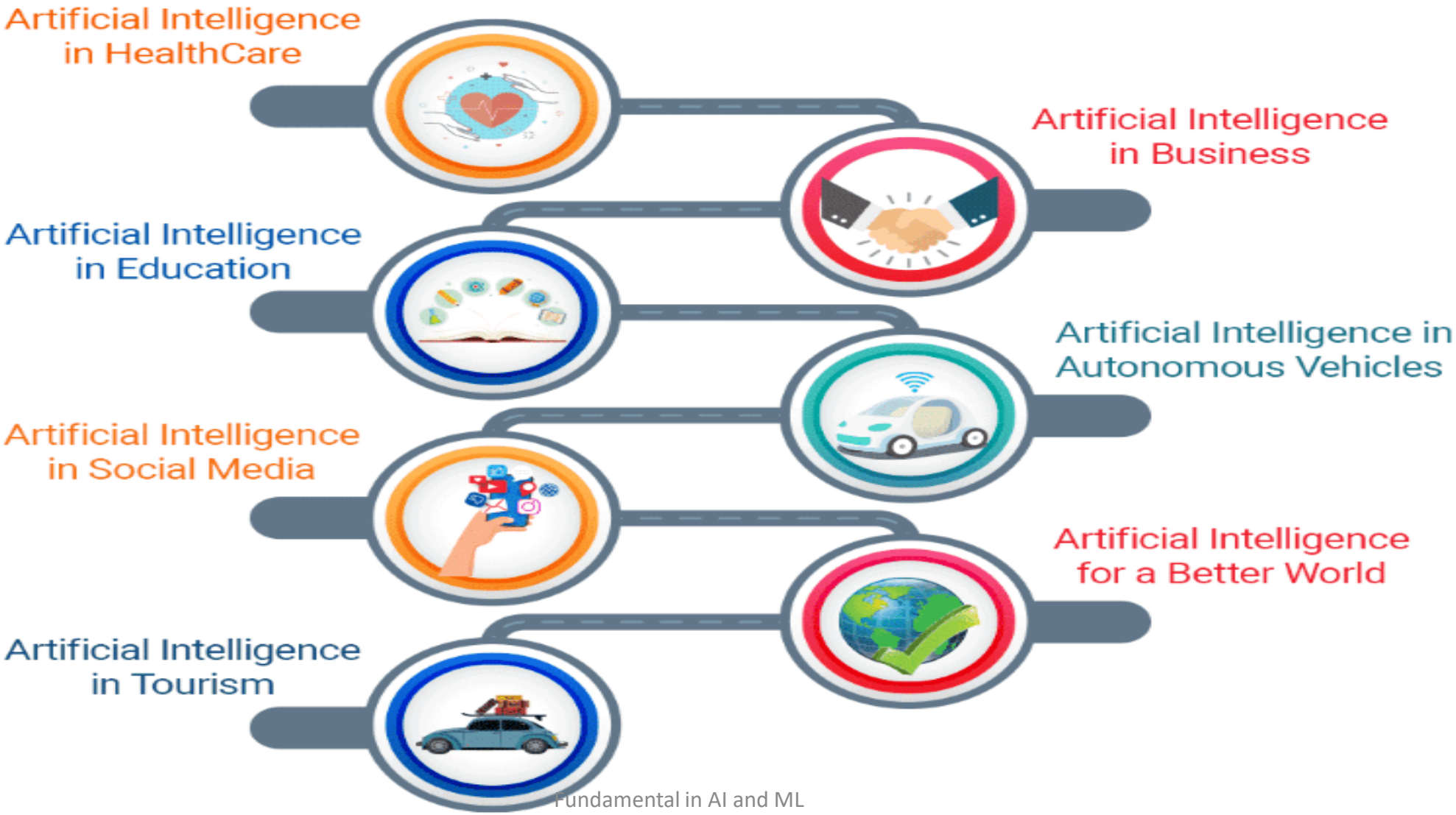
FUNDAMENTALS IN AI AND ML

On the Verge of Major Breakthroughs

Artificial Intelligence (AI) has been moving extremely quickly in the last few years, demonstrating a potential to revolutionize every aspect of our lives



Applications of AI



But, What is AI ?

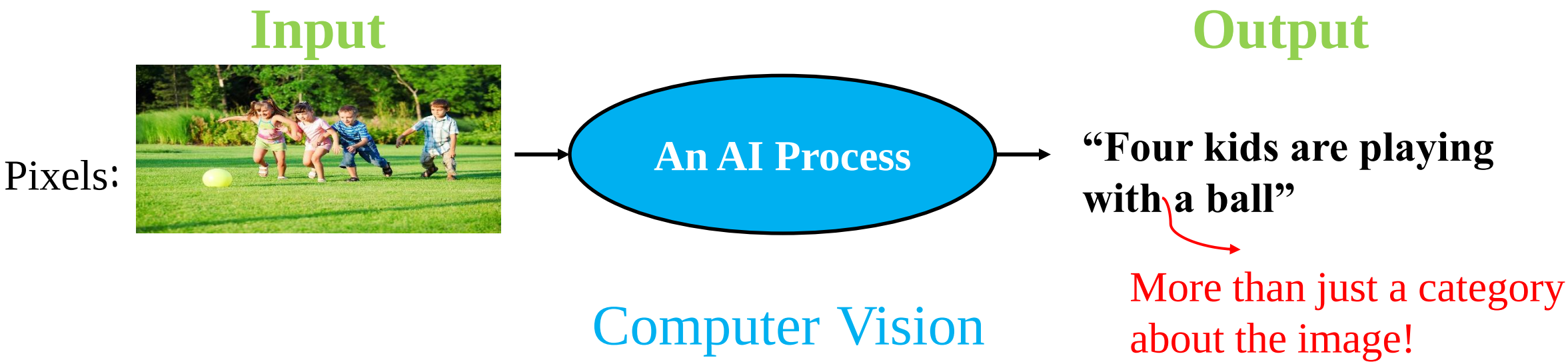
AI can be broadly defined as technology that can *learn* and produce intelligent behavior



Computer Vision

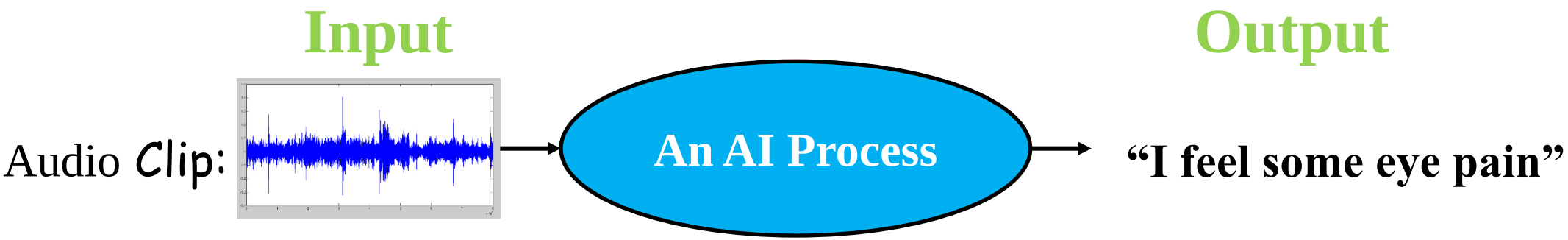
But, What is AI ?

AI can be broadly defined as technology that can *learn* and produce intelligent behavior



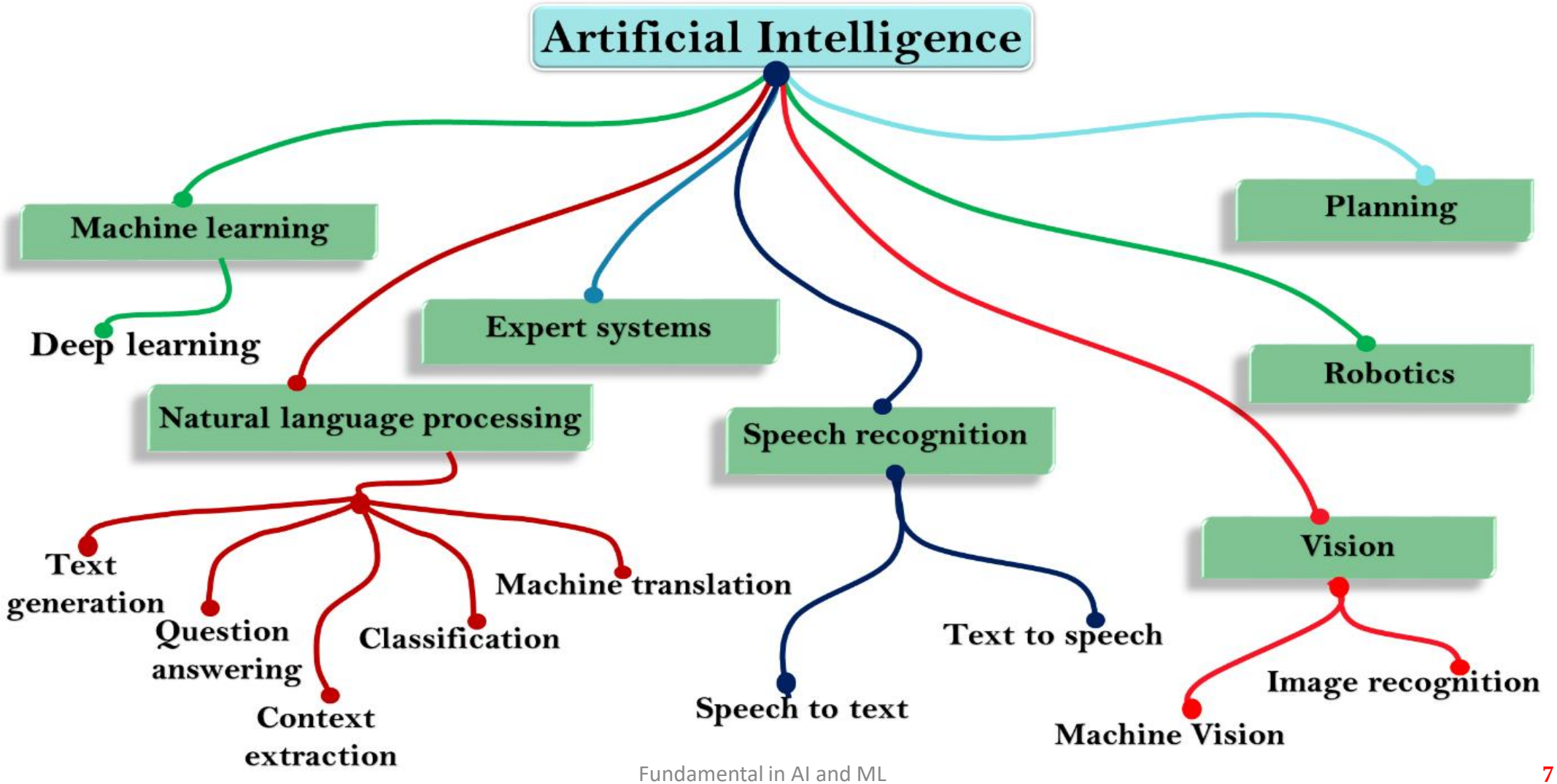
Applications of AI

AI can be broadly defined as technology that can *learn* and produce intelligent behavior

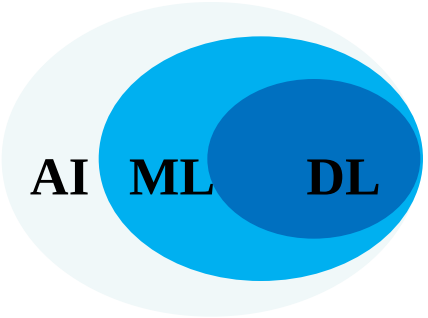


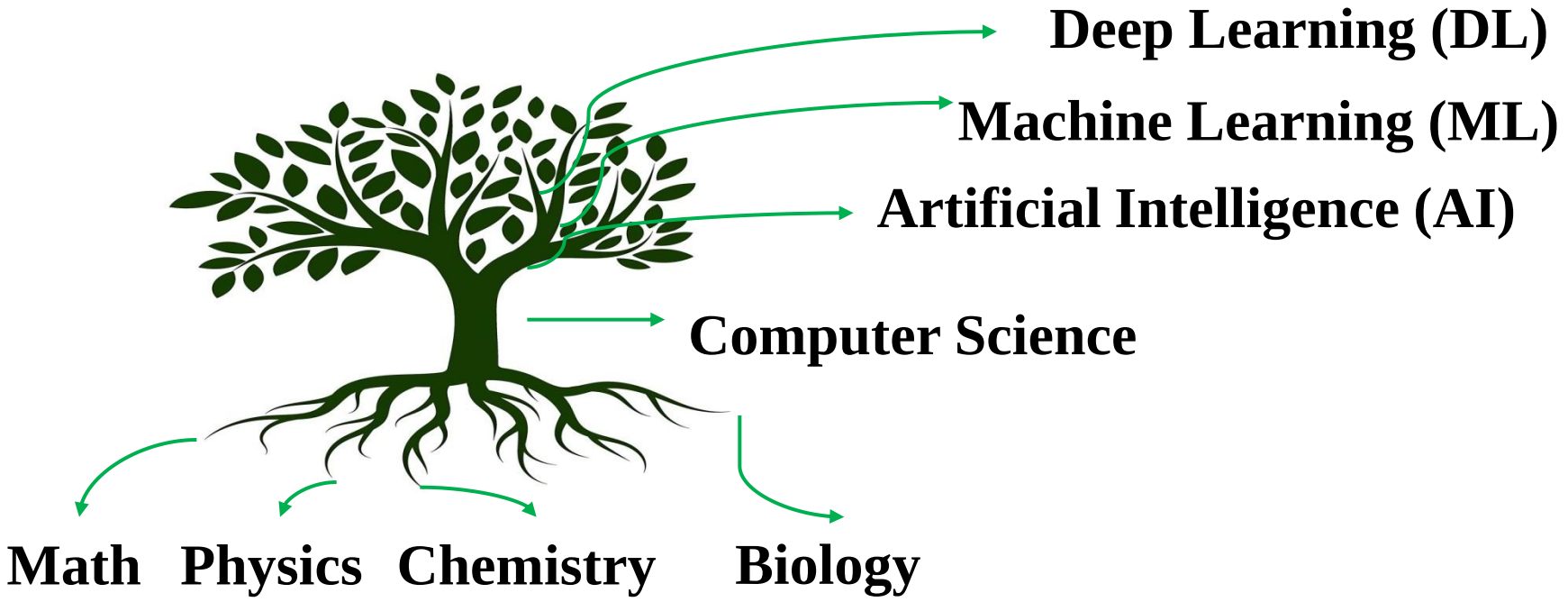
Speech Recognition

Artificial Intelligence



Artificial Intelligence





AI in real time systems

- Search engines (such as Google Search), targeting online advertisements, recommendation systems (offered by Netflix, YouTube or Amazon),
- Driving internet traffic,
- Targeted advertising (AdSense, Facebook),
- Virtual assistants (such as Siri or Alexa),
- Autonomous vehicles (including drones and self-driving cars),
- Automatic language translation (Microsoft Translator, Google Translate),
- Facial recognition (Apple's Face ID or Microsoft's DeepFace),
- Image labeling (used by Facebook, Apple's iPhoto and TikTok) and spam filtering.

Agents in Artificial Intelligence

An AI system can be defined as the study of the rational agent and its environment. The agents sense the environment through sensors and act on their environment through actuators. An AI agent can have mental properties such as knowledge, belief, intention, etc.

What is an Agent?

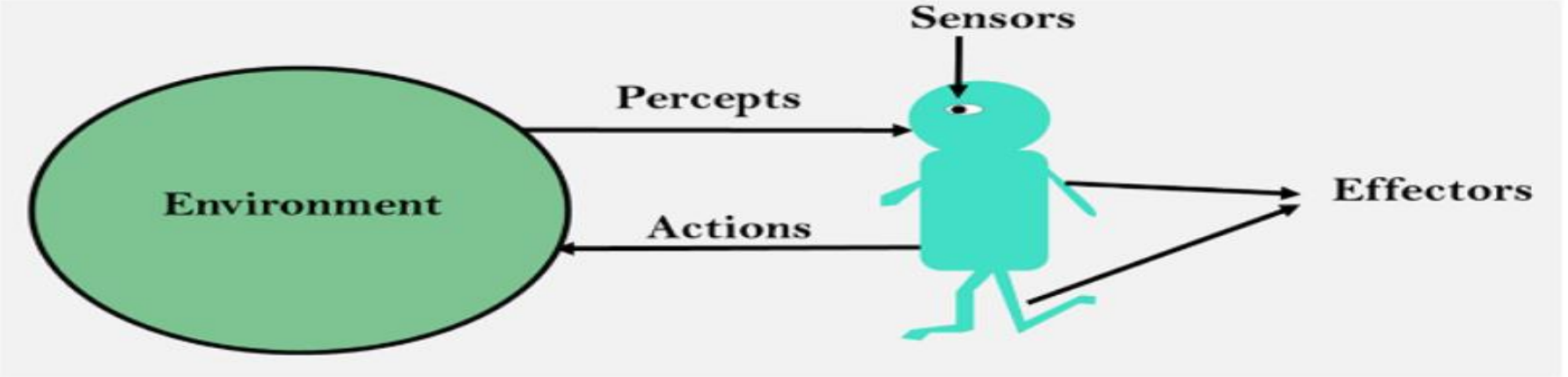
An agent can be anything that perceives the environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**. An agent can be:-

- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

Agents in Artificial Intelligence

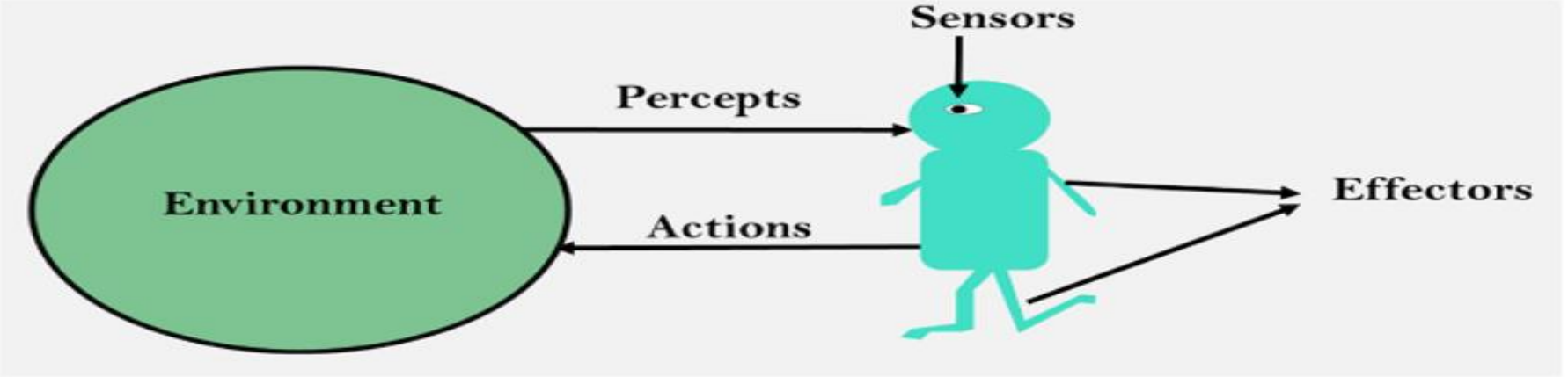
Sensor: Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.



Agents in Artificial Intelligence

Effectors: Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.



Intelligent Agent

An intelligent agent is an autonomous entity that act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI agent:

- **Rule 1:** An AI agent must have the ability to perceive the environment.
- **Rule 2:** The observation must be used to make decisions.
- **Rule 3:** Decision should result in an action.
- **Rule 4:** The action taken by an AI agent must be a rational action.

Rational Agent

- A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.

- A rational agent is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.

- For an AI agent, the rational action is most important because in AI reinforcement learning algorithm, for each best possible action, agent gets the positive reward and for each wrong action, an agent gets a negative reward.

Rationality

The rationality of an agent is measured by its performance measure. Rationality can be judged on the basis of following points:

- Performance measure which defines the success criterion.
- Agent prior knowledge of its environment.
- Best possible actions that an agent can perform.
- The sequence of percepts.

Structure of an AI Agent

The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

Agent = Architecture + Agent program

Architecture: Architecture is machinery that an AI agent executes on.

Agent Function: The agent function is used to map a percept to an action.

$$f:P^* \rightarrow A$$

Agent program: An agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f.

PEAS Representation

PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- **P:** Performance measure
- **E:** Environment
- **A:** Actuators
- **S:** Sensors

Here performance measure is the objective for the success of an agent's behavior.

PEAS for self-driving cars:

Let's suppose a self-driving car then PEAS representation will be:

Performance: Safety, time, legal drive, comfort

Environment: Roads, other vehicles, road signs, pedestrian

Actuators: Steering, accelerator, brake, signal, horn

Sensors: Camera, GPS, speedometer, odometer, accelerometer, sonar.

Example of Agents with their PEAS representation

Agent	Performance Measure	Environment	Actuator	Sensor
Medical Diagnosis	• Healthy Patient • Minimized Cost • Avoid lawsuits,	• Hospital • Staffs • Patients	• Prescription, Diagnosis, • Scan report • questions,tests, diagnoses,treatments,referrals	• keyboard(symptoms, test results, Patient's response)

Example of Agents with their PEAS representation

Agent	Performance Measure	Environment	Actuator	Sensor
Subject Tutoring	• Maximize scores	Classroom, Desk, Chair, Board, Staff, Students	Smart displays, Corrections	Eyes, Ears, Notebooks

Example of Agents with their PEAS representation

Agent	Performance Measure	Environment	Actuator	Sensor
Vacuum Cleaner	• Cleanness • Efficiency • Battery life • Security	• Room • Table • Wood floor • Carpet • Various obstacles	• Wheels • Brushes • Vacuum Extractor	• Camera • Dirt detection sensor • Cliff sensor • Bump Sensor • Infrared Wall Sensor

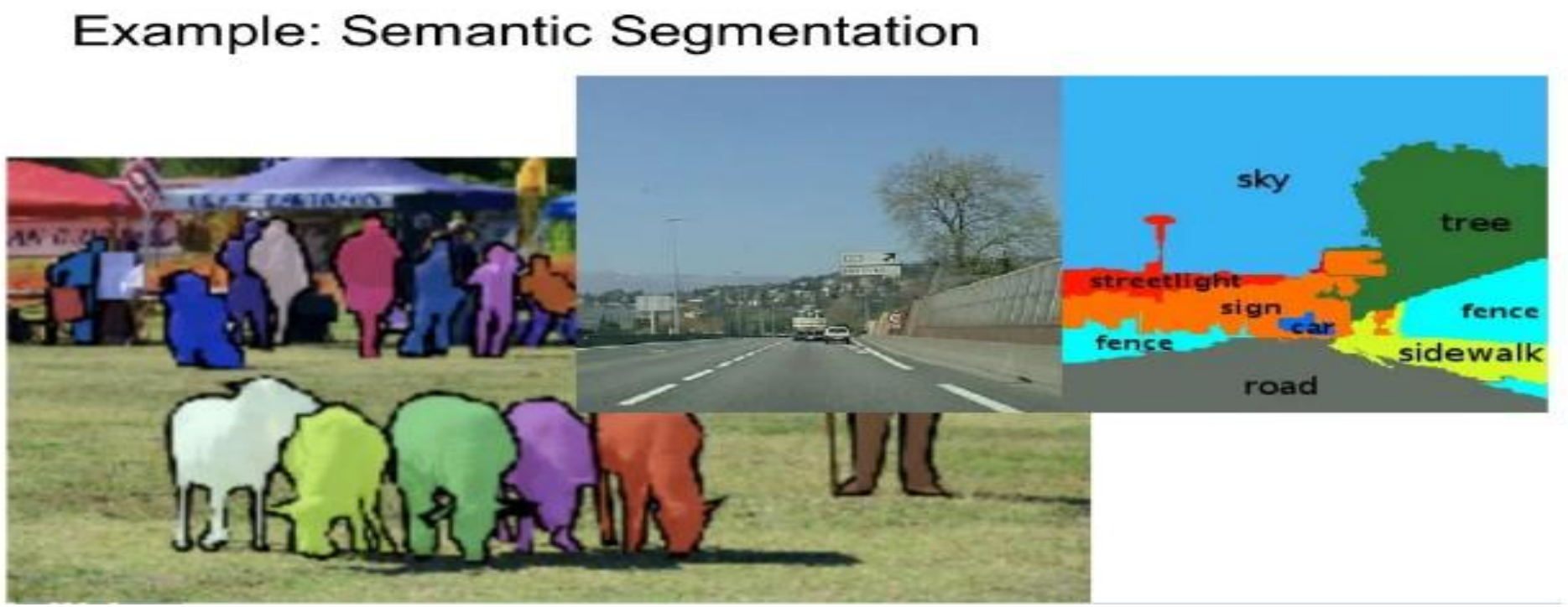
Object Detection



Activity Recognition



Semantic Segmentation



Disease Detection

Example: Radiologist-Level Pneumonia Detection

	F1 Score (95% CI)
Radiologist 1	0.383 (0.309, 0.453)
Radiologist 2	0.356 (0.282, 0.428)
Radiologist 3	0.365 (0.291, 0.435)
Radiologist 4	0.442 (0.390, 0.492)
Radiologist Avg.	0.387 (0.330, 0.442)
CheXNet	0.435 (0.387, 0.481)

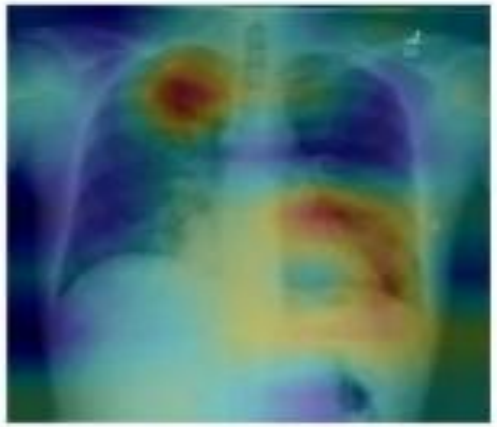
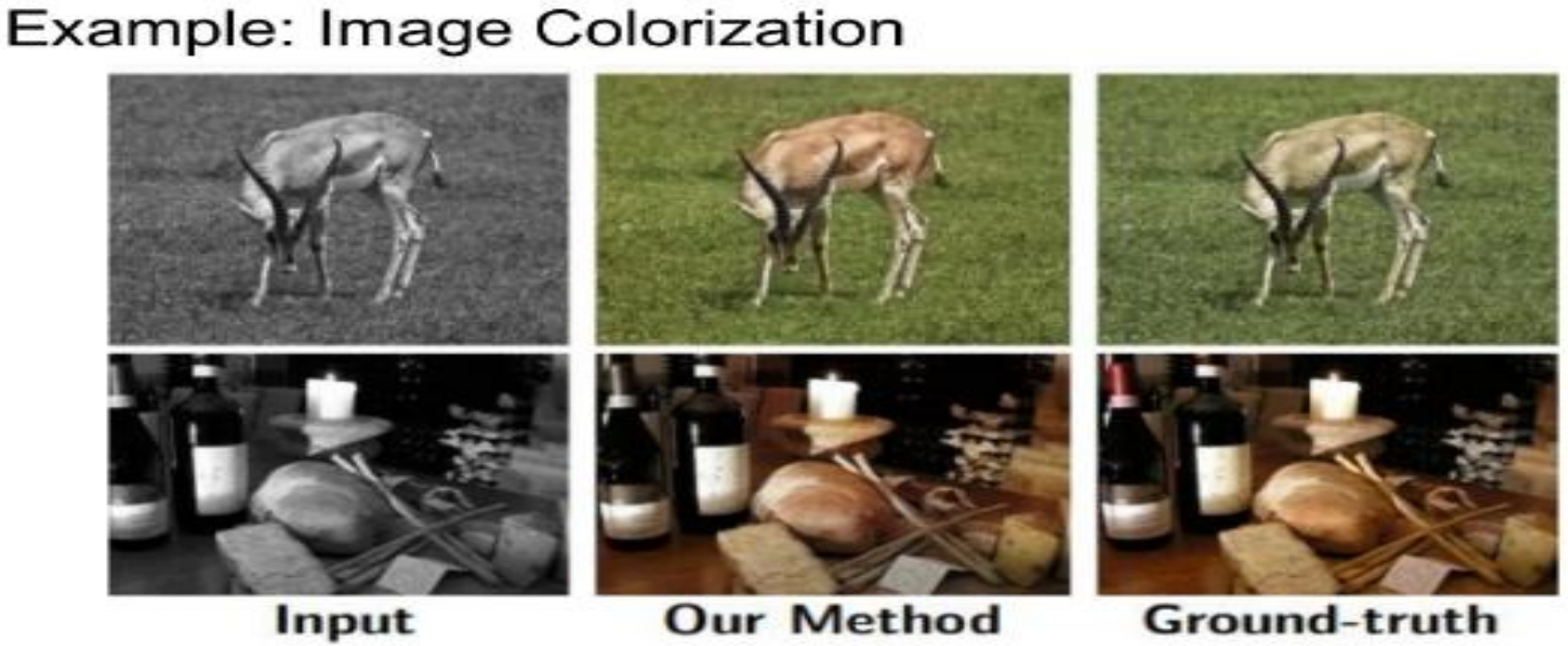


Image colorization



Style Transfer



Style transfer



Lip Sync

Example: Learning Lip Sync from Audio



http://grail.cs.washington.edu/projects/AudioToObama/
https://www.youtube.com/watch?v=9Yq67CjDqvw

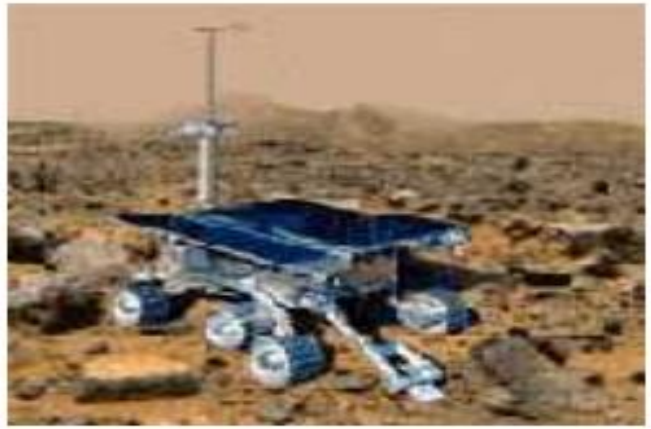
Image to Image Translation



Why interest in AI?



Labor



Science



Appliances

Google

YAHOO!

Search engines



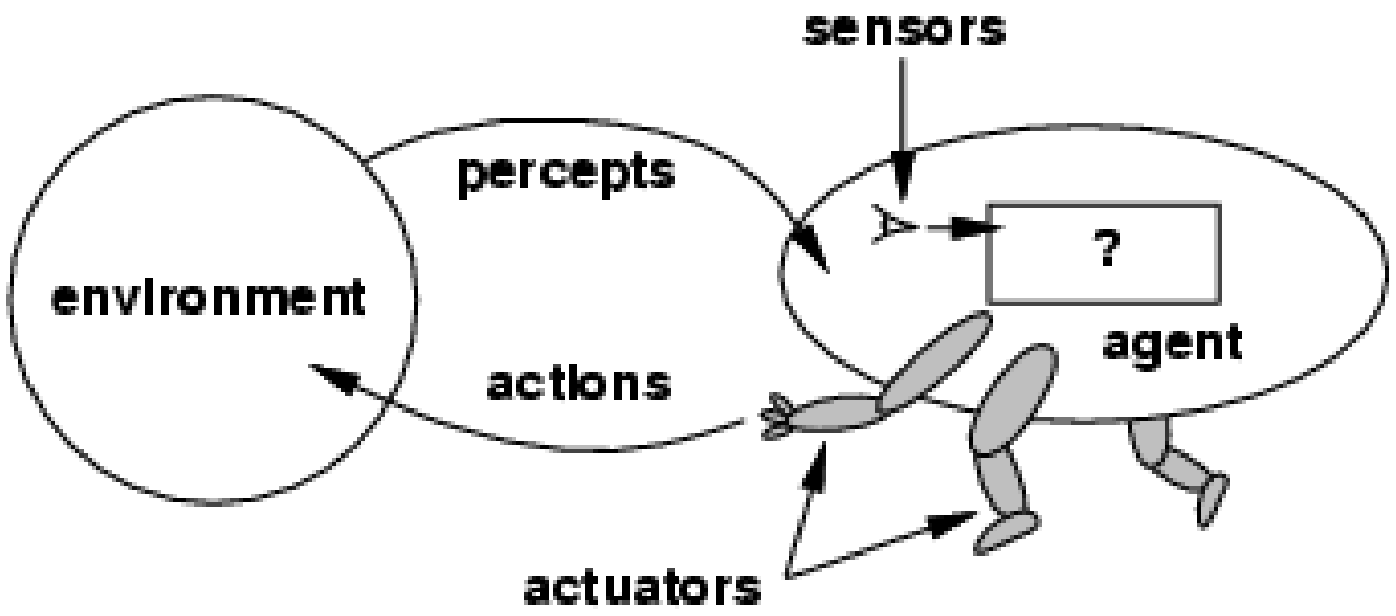
Medicine/
Diagnosis

Activ


AI Agents

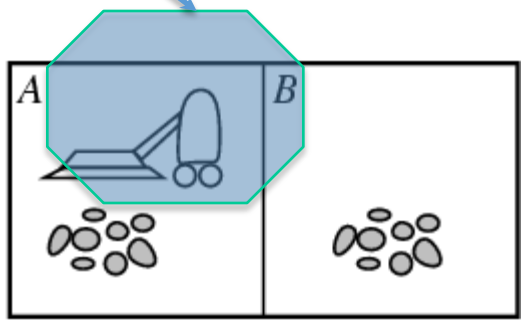
Agents

Definition: An agent perceives its environment via sensors and acts upon that environment through its actuators



E.g., vacuum-cleaner world

Agent / Robot



iRobot Roomba® 400
Vacuum Cleaning Robot



iRobot Corporation

Founder Rodney Brooks (MIT)

Percepts: location and contents

e.g., [A, Dirty]

- Powerful suction and rotating brushes
- Automatically navigates for best cleaning coverage
- Cleans under and around furniture, into corners and along wall edges
- Self-adjusts from carpets to hard floors and back again
- Automatically avoids stairs, drop-offs and off-limit areas
- Simple to use— just press the Clean button and Roomba does the rest

Actions: *Left*, *Right*, *Suck*, *NoOp*

Rational Agents

An agent should strive to "do the right thing", based on what:

- it can perceive and
- the actions it can perform.

The right action is the one that will cause the agent to be most successful

Performance measure of game-playing agent: win/loss percentage (maximize), robustness, unpredictability (to “confuse” opponent), etc.

Rational Agents

Performance measure: An *objective criterion for success* of an agent's behavior.

Performance measures of a vacuum-cleaner agent: amount of dirt cleaned up, amount of time taken, amount of electricity consumed, level of noise generated, etc.

Performance measures self-driving car: time to reach destination (minimize), safety, predictability of behavior for other agents, reliability, etc.

Performance measure of game-playing agent: win/loss percentage (maximize), robustness, unpredictability (to "confuse" opponent), etc.

Rational Agents

For each possible percept sequence, a rational agent should select an action that maximizes its performance measure (in expectation) given the evidence provided by the percept sequence and whatever built- in knowledge the agent has.

Why “in expectation”?

Captures actions with stochastic / uncertain effects or actions performed in stochastic environments. We can then look at the expected value of an action.

In high-risk settings, we may also want to limit the worst-case behavior.

Rational Agents

Notes:

Rationality is distinct from omniscience ("all knowing"). We can behave rationally even when faced with incomplete information.

Agents can perform actions in order to modify future percepts so as to obtain useful information: information gathering, exploration.

An agent is autonomous if its behavior is determined by its own experience (with ability to learn and adapt).

Characterizing a Task Environment

Must first specify the setting for intelligent agent design.

PEAS: Performance measure, Environment, Actuators, Sensors

Example: the task of designing a self-driving car



- Performance measure Safe, fast, legal, comfortable trip
- Environment Roads, other traffic, pedestrians
- Actuators Steering wheel, accelerator, brake, signal, horn
- Sensors Cameras, LIDAR (light/radar), speedometer, GPS, odometer, engine sensors, keyboard

PEAS Examples

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Task Environments

1) Fully observable / Partially observable

If an agent's sensors give it access to the complete state of the environment needed to choose an action, the environment is fully observable.

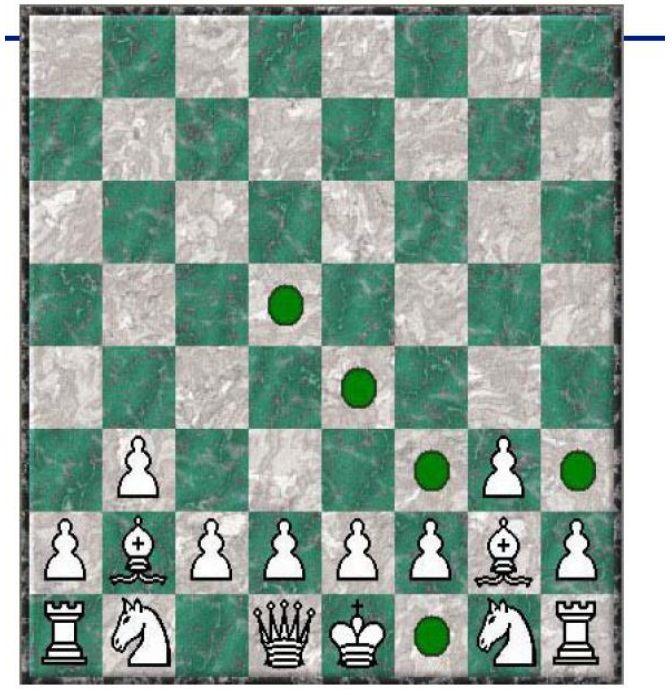
(e.g. chess – what about Kriegspiel?)



Task Environments

Making things a bit more challenging…

***Kriegspiel* --- you can't see your opponent!**



Incomplete / uncertain information inherent in the game.

Balance exploitation (best move given current knowledge) and exploration (moves to explore where opponent's pieces might be).

Use probabilistic reasoning techniques.

Task Environments

2) Deterministic / Stochastic

- An environment is **deterministic** if the next state of the environment is completely determined by the current state of the environment and the action of the agent;
- In a **stochastic** environment, there are multiple, unpredictable outcomes. (If the environment is deterministic except for the actions of other agents, then the environment is **strategic**).

In a fully observable, deterministic environment, the agent need not deal with uncertainty.

Note: Uncertainty can also arise because of computational limitations. E.g., we may be playing an omniscient ("all knowing") opponent but we may not be able to compute his/her moves.

Task Environments

3) Episodic / Sequential

- **In an episodic environment, the agent's experience is divided into atomic episodes. Each episode consists of the agent perceiving and then performing a single action.**
- **Subsequent episodes do not depend on what actions occurred in previous episodes. Choice of action in each episode depends only on the episode itself. (E.g., classifying images.)**
- **In a sequential environment, the agent engages in a series of connected episodes. Current decision can affect future decisions. (E.g., chess and driving)**

Task Environments

4) Static / Dynamic

- **A static environment does not change while the agent is thinking.**
- **The passage of time as an agent deliberates is irrelevant.**
- **The environment is semidynamic if the environment itself does not change with the passage of time but the agent's performance score does.**

Task Environments

5) Discrete / Continuous

- **If the number of distinct percepts and actions is limited, the environment is discrete, otherwise it is continuous.**

6) Single agent / Multi-agent

- **If the environment contains other intelligent agents, the agent needs to be concerned about strategic, game-theoretic aspects of the environment (for either cooperative *or* competitive agents).**
- **Most engineering environments don't have multi-agent properties, whereas most social and economic systems get their complexity from the interactions of (more or less) rational agents.**

Example Tasks and Environment Types

	Chess with a clock	Chess without a clock	Taxi driving
Fully observable	Yes	Yes	No
Deterministic	Strategic	Strategic	No
Episodic	No	No	No
Static	Semi	Yes	No
Discrete	Yes	Yes	No
Single agent	No	No	No

The environment type largely determines the agent design

The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

How to make the right decisions? **Decision theory**

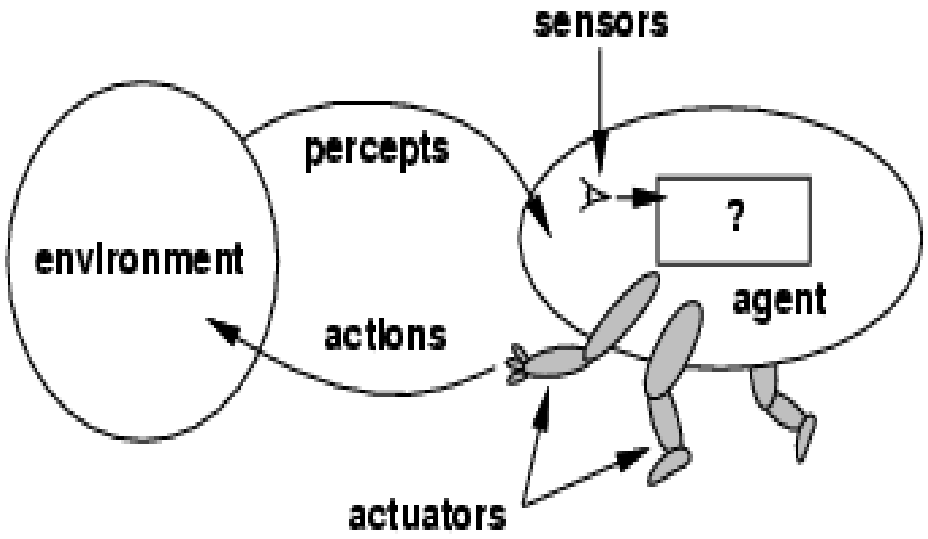
Task Environment

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Exercise on Task environment

- Playing soccer.
- Exploring the subsurface oceans of Titan.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.
- Performing a high jump.
- Knitting a sweater.
- Bidding on an item at an auction.

Agents and Environment



The agent function maps from percept histories to actions

The agent program runs (internally) on the physical architecture to produce *f*

agent = architecture + program


Types of Agents

I) Table-lookup driven agents

Uses a percept sequence / action table in memory to

find the next action. Implemented as a (large) lookup table.

- Drawbacks:
  - Huge table (often simply too large)
  - Takes a long time to build/learn the table

I) Table-lookup driven agents

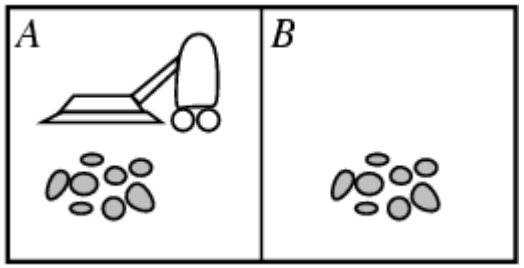
Toy example:Vacuum world.

Percepts: robot senses it's location and "cleanliness."

So, location and contents, e.g., [A, Dirty], [B, Clean].

With 2 locations, we get 4 different possible sensor inputs.

Actions: *Left*, *Right*, *Suck*, *NoOp*



Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Table driven agent

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions, indexed by percept sequences, initially fully specified

  append percept to the end of percepts
  action ← LOOKUP(percepts, table)
  return action
```

Table LookupII) --- Simple reflex agents

So, table lookup formulation is mainly of theoretical interest. For practical agent systems, we need to find much more compact representations.

For example, logic-based representations, Bayesian net representations, or neural net style representations, or use a different agent architecture,

e.g., "ignore the past" --- Reflex agents.

II) Simple reflex agents

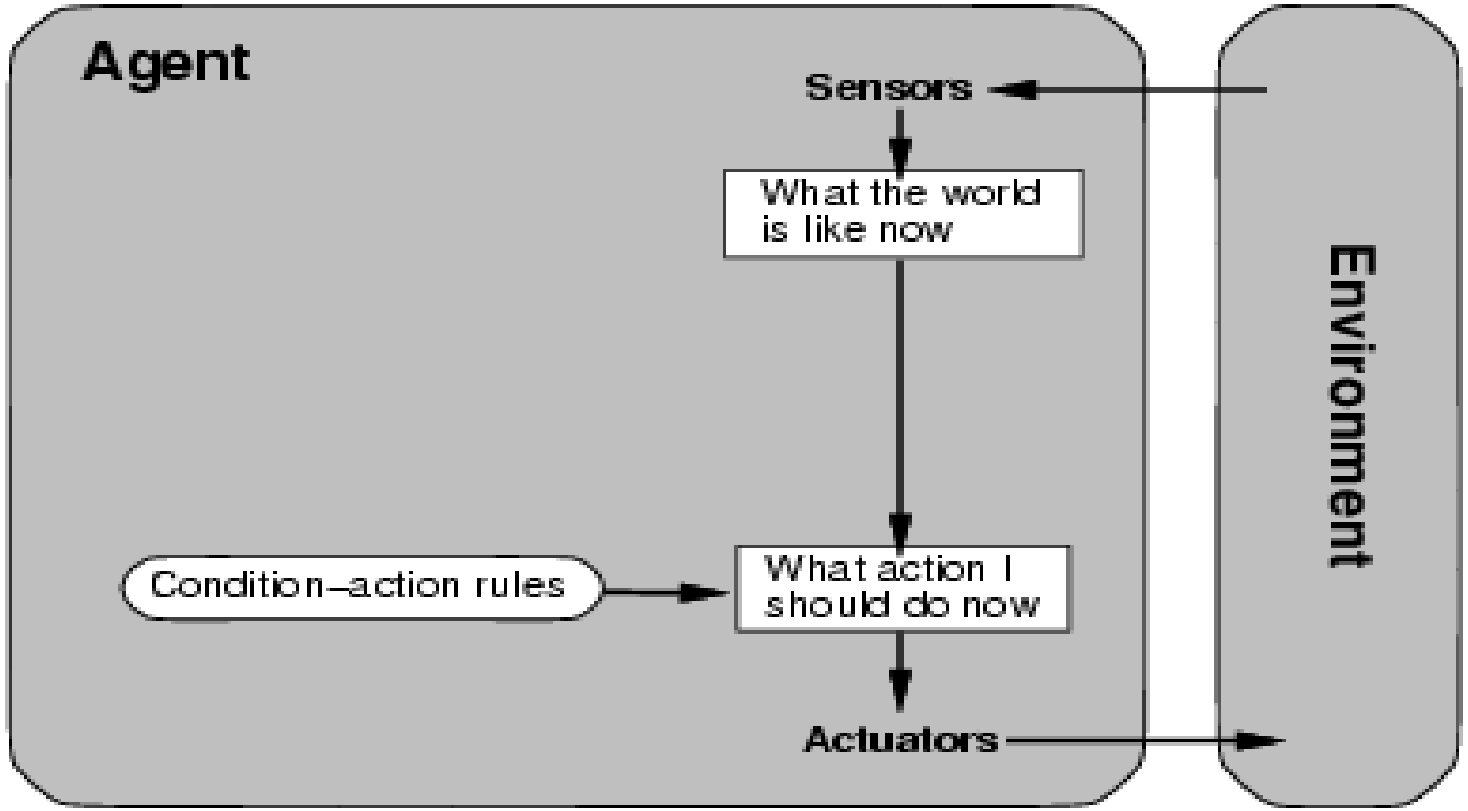
Agents **do not have memory** of past world states or percepts.

So, actions depend solely on **current percept**. Action becomes a “reflex.”

Uses **condition-action rules**.

II) Simple reflex agents

Agent selects actions on the basis of *current* percept only.



If tail-light of car in front is red, then brake.

II) Simple reflex agents

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
   persistent: rules, a set of condition–action rules

   state ← INTERPRET-INPUT(percept)
   rule ← RULE-MATCH(state, rules)
   action ← rule.ACTION
   return action
```

II) Simple reflex agents

Closely related to “behaviorism” (psychology; quite effective in explaining lower-level animal behaviors, such as the behavior of ants and mice).

The Roomba robot largely behaves like this. Behaviors are robust and can be quite effective and surprisingly complex.

But, how does complex behavior arise from simple reflex behavior?
E.g. ants colonies and bee hives are quite complex.

Simple rules in a diverse environment can give rise to surprising complexity.

See A-life work (artificial life) community, and Wolfram’s cellular automata.

III) --- Model-based reflex agents

Key difference (wrt simple reflex agents):

- Agents have **internal state**, which is used to keep track of past states of the world.
- Agents have the ability **to represent change in the World**.

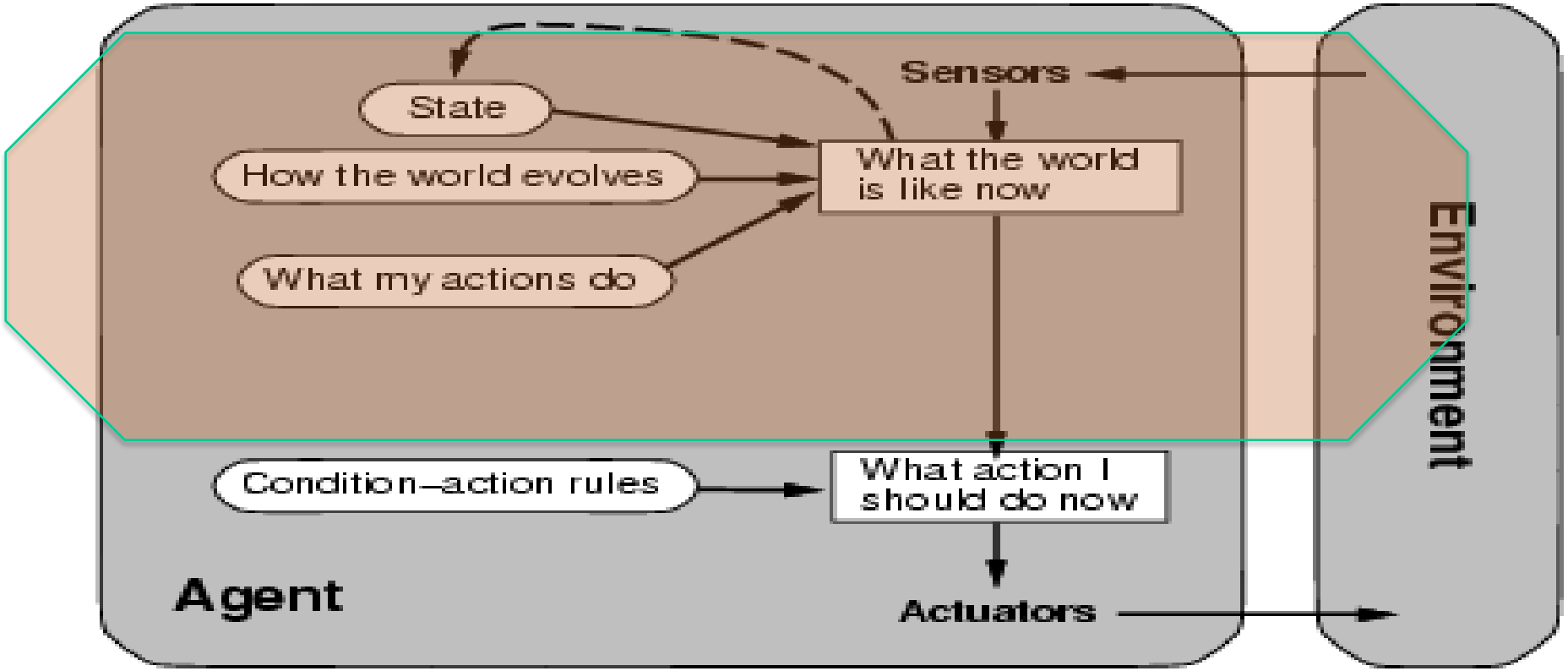
Example: Rodney Brooks' Subsumption Architecture

--- behavior based robots.

III) --- Model-based reflex agents

Module:
Logical Agents
Representation and Reasoning: Part III/IV R&N

How detailed?



If "dangerous driver in front,"
then "keep distance."

"Infers potentially dangerous driver in front."

III) --- Model-based agents

```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
              model, a description of how the next state depends on current state and action
              rules, a set of condition–action rules
              action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

III) --- Model-based agents An example: Brooks' Subsumption Architecture

Main idea: build complex, intelligent robots by decomposing behaviors into a hierarchy of skills, each defining a percept-action cycle for one very specific task.

Examples: collision avoidance, wandering, exploring, recognizing doorways, etc.

Each behavior is modeled by a finite-state machine with a few states (though each state may correspond to a complex function or module; provides internal state to the agent).

Behaviors are loosely coupled via asynchronous interactions.

Note: minimal internal state representation. p. 1003 R&N

III) --- Model-based agents An example: Brooks' Subsumption Architecture

In subsumption architecture, increasingly complex behaviors arise from the combination of simple behaviors.

The most basic simple behaviors are on the level of reflexes: • avoid an object; • go toward food if hungry, • move randomly.

A more complex behavior that sits on top of simple behaviors may be "go across the room."

The more complex behaviors subsume the less complex ones to accomplish their goal.

How much of an internal model of the world?

Planning in and reasoning about our surroundings appears to require some kind of internal representation of our world.

We can "try" things out in this representation. Much like running a "simulation" of the effect of actions or a sequence of actions in our head.

General assumption for many years:

The more detailed internal model, the better.

IV) Goal-based agents

Key difference wrt Model-Based Agents:

In addition to state information, have **goal information** that **describes desirable situations to be achieved**.

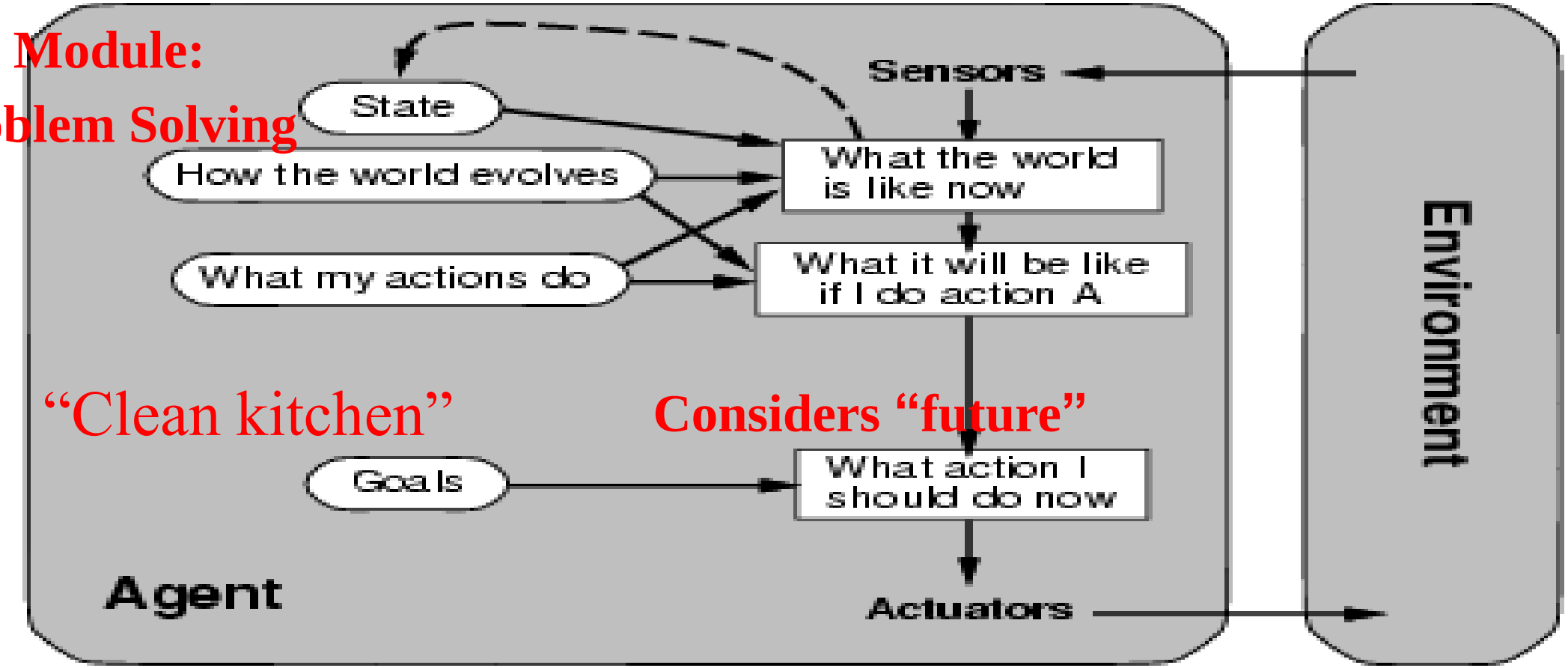
Agents of this kind take future events into consideration.

What *sequence* of actions can I take to achieve certain goals?

Choose actions so as to (eventually) achieve a (given or computed) goal.

- ***problem solving and search! (R&N --- Part II, chapters 3 to 6)***

IV) Goal-based agents



Agent keeps track of the world state as well as set of goals it's trying to achieve: chooses actions that will (eventually) lead to the goal(s).

More flexible than reflex agents may involve search and planning

V) Utility-based agents

When there are multiple possible alternatives, how to decide which one is best?

Goals are qualitative: A goal specifies a crude(not carefully) distinction between a happy and unhappy state, but often need a more general performance measure that describes "degree of happiness."

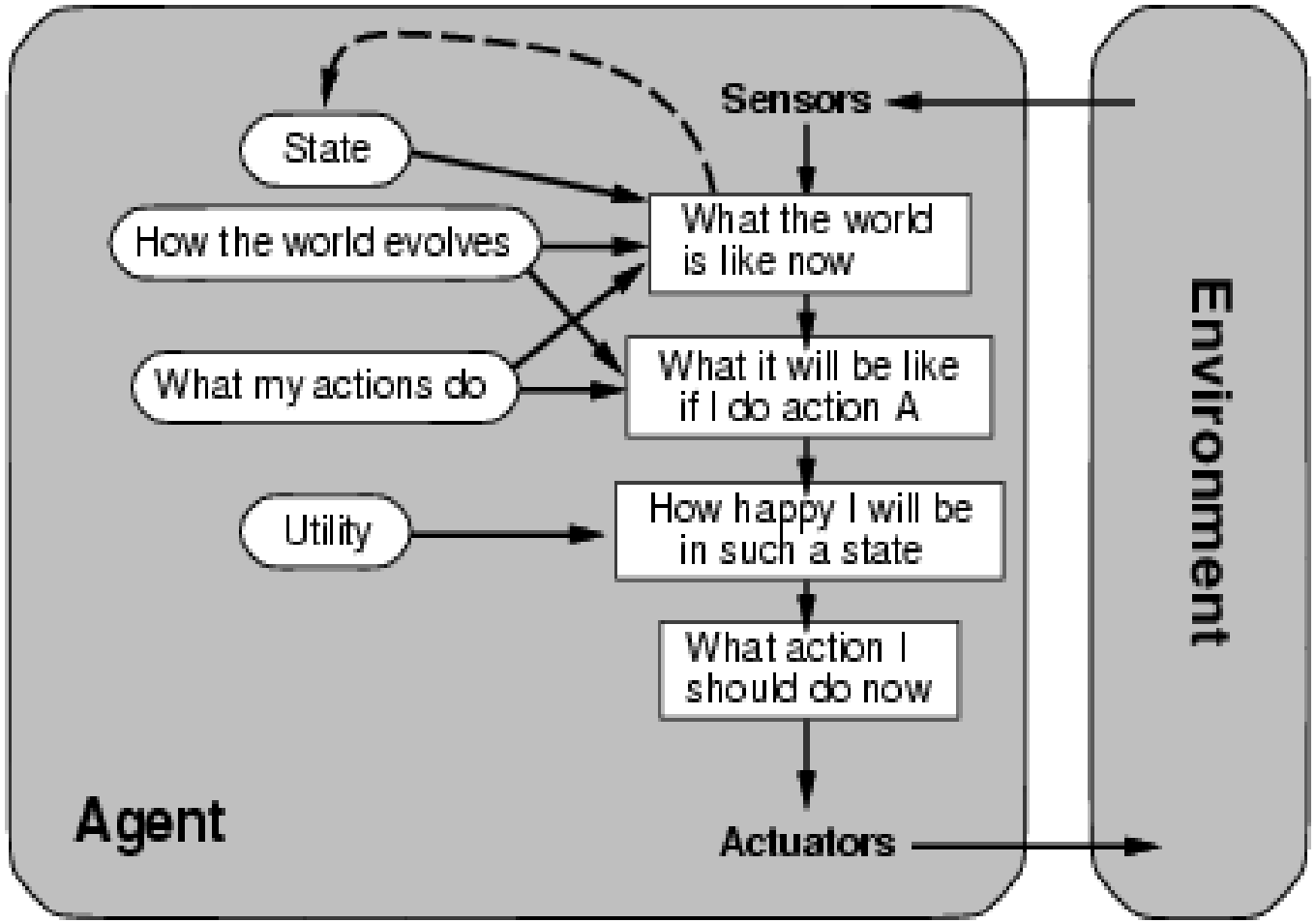
Utility function U: State → R indicating a measure of success or happiness when at a given state.

Important for making tradeoffs: Allows decisions comparing choice between conflicting goals, and choice between likelihood of success and importance of goal (if achievement is uncertain).

Use decision theoretic models: e.g., faster vs. safer.

V) Utility-based agents VI) --- Learning agents
Adapt and improve over time

Module:
Decision Making

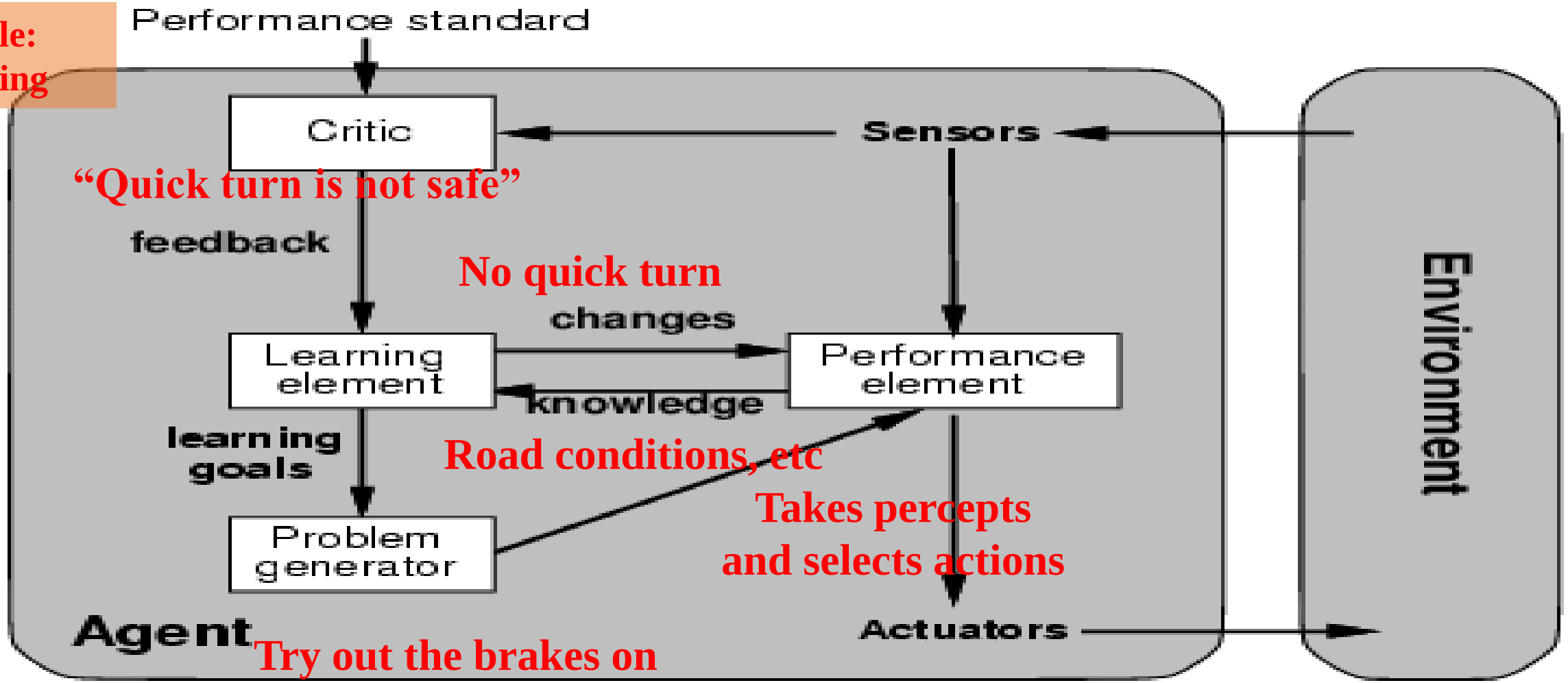


Decision theoretic actions:
e.g. faster vs. safer

VI) --- Learning agents Adapt and improve over time

More complicated when agent needs to learn utility information: Reinforcement learning (based on action payoff)

Module: Learning



Summary: Agent Types

(1) Table-driven agents

- **use a percept sequence/action table in memory to find the next action. They are implemented by a (large) lookup table.**

(2) Simple reflex agents

- **are based on condition-action rules, implemented with an appropriate production system. They are stateless devices which do not have memory of past world states.**

(3) Agents with memory - Model-based reflex agents

- **have internal state, which is used to keep track of past states of the world.**

Summary: Agent Types

(4) Agents with goals – Goal-based agents

- **are agents that, in addition to state information, have goal information that describes desirable situations. Agents of this kind take future events into consideration.**

(5) Utility-based agents

- **base their decisions on classic axiomatic utility theory in order to act rationally.**

(6) Learning agents

- **they have the ability to improve performance through learning.**

Summary: Agent Types

- **An agent perceives and acts in an environment, has an architecture, and is implemented by an agent program.**
- **A rational agent always chooses the action which maximizes its expected performance, given its percept sequence so far.**
- **An autonomous agent uses its own experience rather than built-in knowledge of the environment by the designer.**

Summary: Agent Types

- **An agent program maps from percept to action and updates its internal state.**
  - **Reflex agents (simple / model-based) respond immediately to percepts.**
  - **Goal-based agents act in order to achieve their goal(s), possible sequence of steps.**
  - **Utility-based agents maximize their own utility function.**
  - **Learning agents improve their performance through learning.**
- **Representing knowledge is important for successful agent design.**
- **The most challenging environments are partially observable, stochastic, sequential, dynamic, and continuous, and contain multiple intelligent agents.**

Module 1
Fundamentals in AI and ML

Supporting Material

What is Artificial Intelligence?

* In today's world, technology is growing very fast, and we are getting in touch with different new technologies day by day.

* Here, one of the booming technologies of computer science is Artificial Intelligence which is ready to create a new revolution in the world by making intelligent machines.

* The Artificial Intelligence is now all around us. It is currently working with a variety of subfields, ranging from general to specific, such as self-driving cars, playing chess, proving theorems, playing music, Painting, etc.

What is Artificial Intelligence?

* AI is one of the fascinating and universal fields of Computer science which has a great scope in future.

* AI holds a tendency to cause a machine to work as a human.

* Artificial Intelligence is composed of two words **Artificial** and **Intelligence**, where Artificial defines *"man-made,"* and intelligence defines *"thinking power"*, hence AI means *"a man-made thinking power.”*

What is Artificial Intelligence?

* we can define AI as:

  "It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions.“

* Artificial Intelligence exists when a machine can have human based skills such as learning, reasoning, and solving problems.

Define by other Scientist

* (Bellman, 1978) defined AI as:

  "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..."

* (Haugeland, 1985) defined AI as:

  “The exciting new effort to make computers think . . . machines with minds, in the full and literal sense"

Define by other Scientist

* (Charniak and McDermott, 1985) defined AI as:

  "The study of mental faculties through the use of computational models"

* (Kurzweil, 1990) defined AI as:

  “The art of creating machines that perform functions that require intelligence when performed by people”

Define by other Scientist

* (Schalkoff, 1 990) defined AI as:

  “A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes"

* (Rich and Knight, 1991) defined AI as:

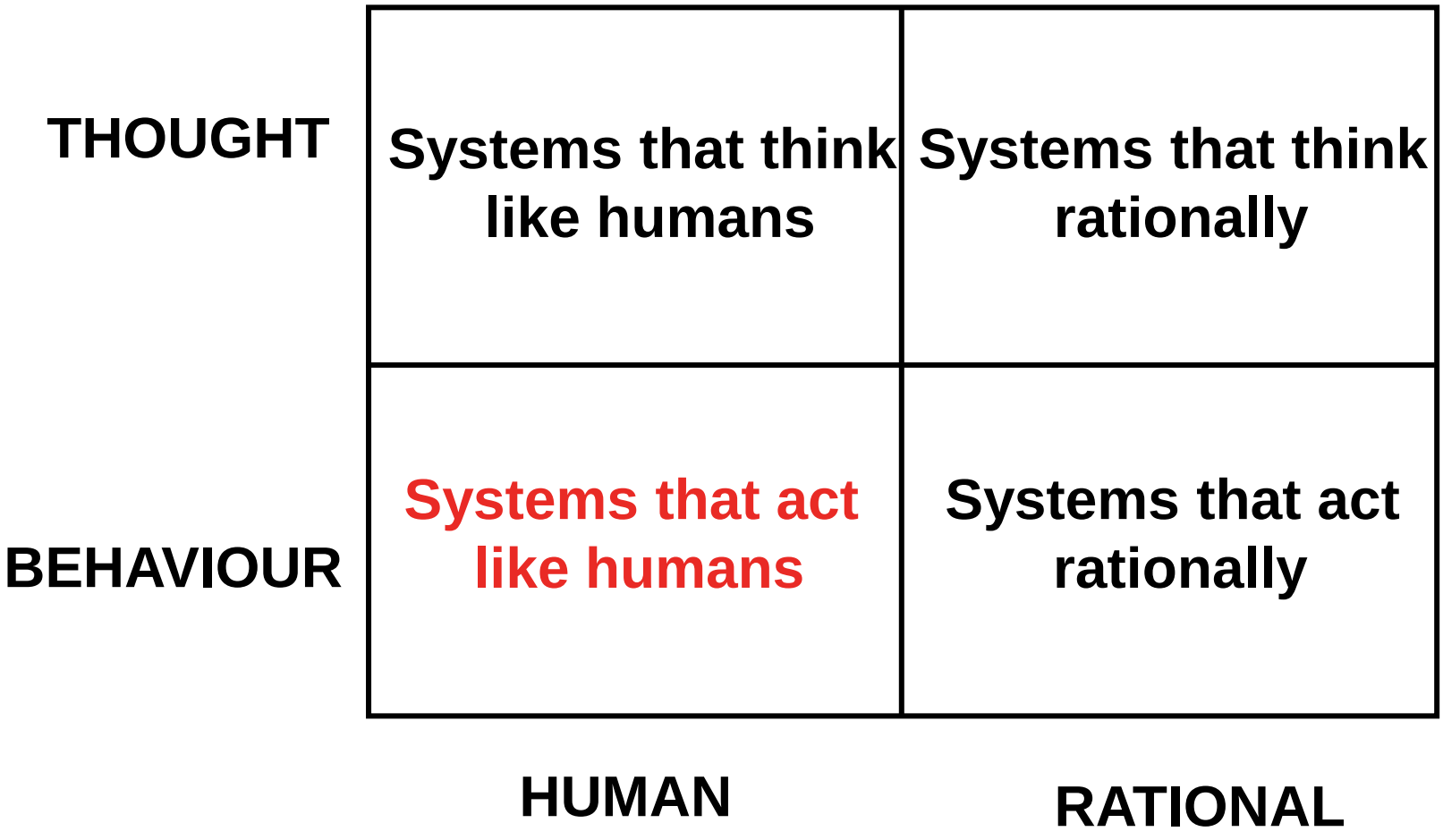
  "The study of how to make computers do things at which, at the moment, people are Better”

Define by other Scientist

* (Luger and Stubblefield, 1993) defined AI as:

  "The branch of computer science that is concerned with the automation of intelligent behavior"

What is Artificial Intelligence ?

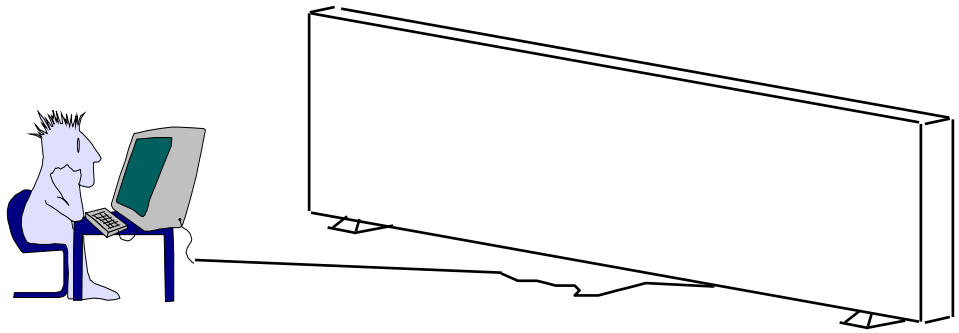


Systems that act like humans: Turing Test

- “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil)

- “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight)

Systems that act like humans



?

- You enter a room which has a computer terminal. You have a fixed period of time to type what you want into the terminal, and study the replies. At the other end of the line is either a human being or a computer system.

- If it is a computer system, and at the end of the period you cannot reliably determine whether it is a system or a human, then the system is deemed to be intelligent.

Systems that act like humans

- The Turing Test approach
  - a human questioner cannot tell if
    - there is a computer or a human answering his question, via teletype (remote communication)
  - The computer must behave intelligently
- Intelligent behavior
  - to achieve human-level performance in all cognitive tasks

Systems that act like humans

- These cognitive tasks include:
  - *Natural language processing*
    - for communication with human
  - *Knowledge representation*
    - to store information effectively & efficiently
  - *Automated reasoning*
    - to retrieve & answer questions using the stored information
  - *Machine learning*
    - to adapt to new circumstances

The total Turing Test

Includes two more issues:

Computer vision

to perceive objects (seeing)

Robotics

to move objects (acting)

What is Artificial Intelligence ?

	HUMAN	RATIONAL
THOUGHT	**Systems that think like humans**	**Systems that think rationally**
BEHAVIOUR	**Systems that act like humans**	**Systems that act rationally**

Systems that think like humans: cognitive modeling

Humans as observed from 'inside'

How do we know how humans think?

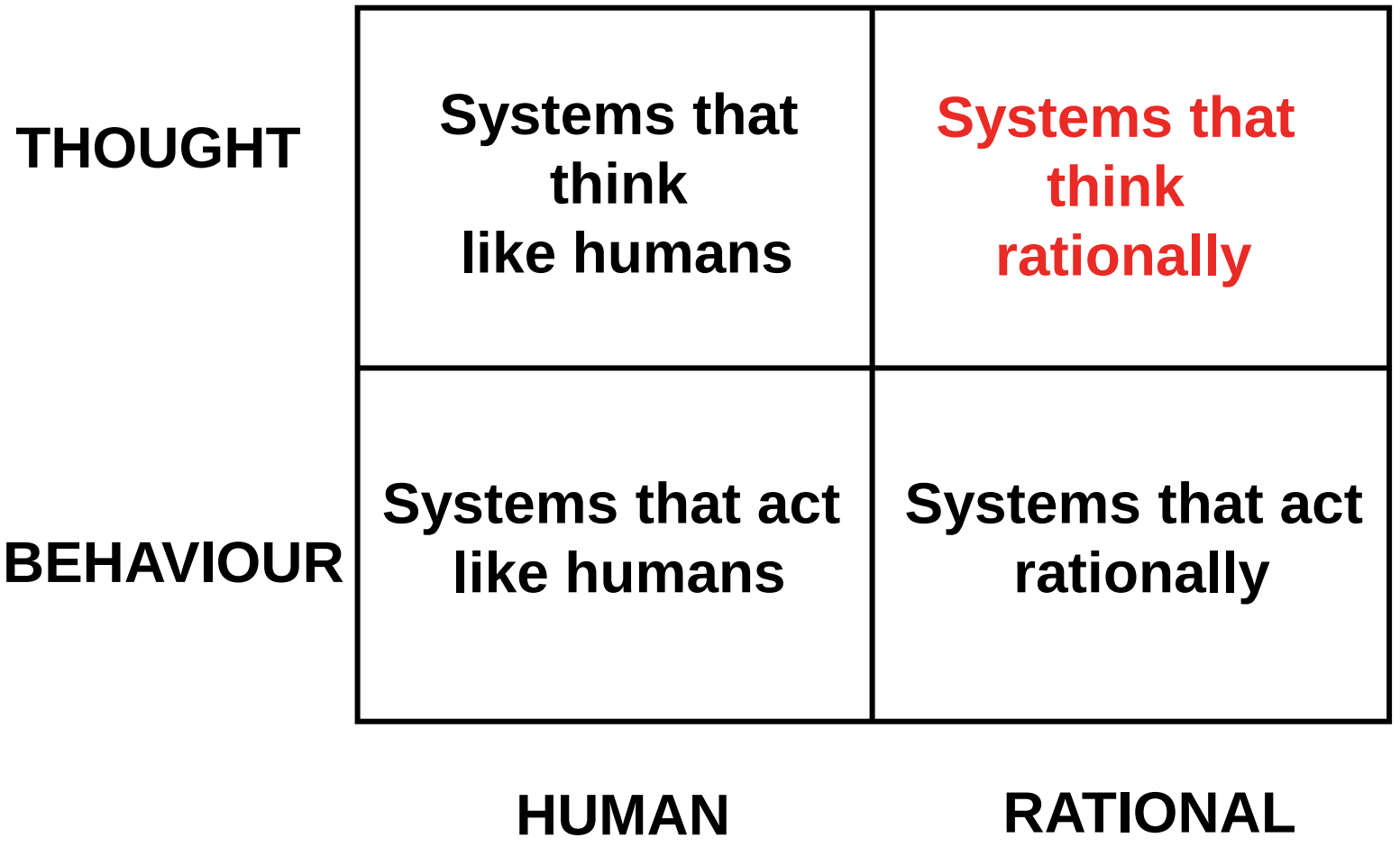
Introspection vs. psychological experiments

Cognitive Science

"The exciting new effort to make computers think … machines with *minds* in the full and literal sense" (Haugeland)

"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning …" (Bellman)

What is Artificial Intelligence ?



Systems that think ‘rationally’ "laws of thought"

Humans are not always ‘rational’

Rational - defined in terms of logic?

Logic can’t express everything (e.g. uncertainty)

Logical approach is often not feasible in terms of computation time (needs ‘guidance’)

“The study of mental facilities through the use of computational models” (Charniak and McDermott)

“The study of the computations that make it possible to perceive, reason, and act” (Winston)

What is Artificial Intelligence ?

	HUMAN	RATIONAL
THOUGHT	**Systems that think like humans**	**Systems that think rationally**
BEHAVIOUR	**Systems that act like humans**	**Systems that act rationally**

Systems that act rationally: “Rational agent”

- Rational behavior: doing the right thing
- The right thing: that which is expected to maximize goal achievement, given the available information
- Giving answers to questions is ‘acting’.
- I don't care whether a system:
  - replicates human thought processes
  - makes the same decisions as humans
  - uses purely logical reasoning

Systems that act rationally

Logic → only *part* of a rational agent, not *all* of rationality

Sometimes logic cannot reason a correct conclusion

At that time, some *specific (in domain) human knowledge* or information is used

Thus, it covers more generally different situations of problems

Compensate the incorrectly reasoned conclusion

Systems that act rationally

Study AI as rational agent –

2 advantages:

It is more general than using logic only

Because: LOGIC + Domain knowledge

It allows extension of the approach with more scientific methodologies

Rational agents

- An agent is an entity that perceives and acts
- This course is about designing rational agents
- Abstractly, an agent is a function from percept histories to actions:
- [*f*: P* → A]
- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Caveat: computational limitations make perfect rationality unachievable
- → design best program for given machine resources

Artificial

Produced by human art or effort, rather than originating naturally.

Intelligence is the ability to acquire knowledge and use it" [Pigford and Baur]

So AI was defined as:

AI is the study of ideas that enable computers to be intelligent.

AI is the part of computer science concerned with design of computer systems that exhibit human intelligence(From the Concise Oxford Dictionary)

From the above two definitions, we can see that AI has two major roles:

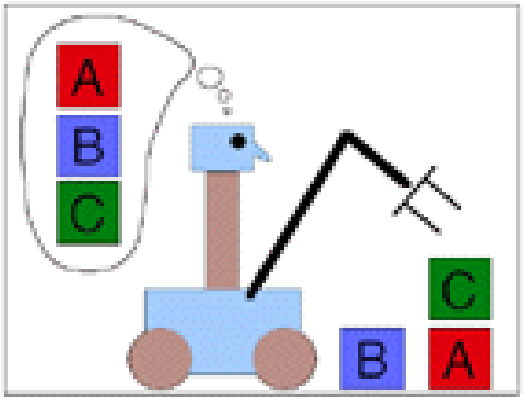
Study the intelligent part concerned with humans.

Represent those actions using computers.

Goals of AI

To make computers more useful by letting them take over dangerous or tedious tasks from human

Understand principles of human intelligence



The Foundation of AI

Philosophy

At that time, the study of human intelligence began with no formal expression

Initiate the idea of mind as a machine and its internal operations

The Foundation of AI

- Mathematics formalizes the three main area of AI: *computation*, *logic*, and *probability*
- Computation leads to analysis of the problems that can be computed
  - *complexity theory*
- Probability contributes the *"degree of belief"* to handle *uncertainty* in AI
- *Decision theory* combines *probability theory* and *utility theory* (bias)

The Foundation of AI

Psychology

How do humans think and act?

The study of human reasoning and acting

Provides reasoning models for AI

Strengthen the ideas

humans and other animals can be considered as information processing machines

The Foundation of AI

Computer Engineering

How to build an efficient computer?

Provides the artifact that makes AI application possible

The power of computer makes computation of large and difficult problems more easily

AI has also contributed its own work to computer science, including: time-sharing, the linked list data type, OOP, etc.

Control theory and Cybernetics

How can artifacts operate under their own control?

The artifacts adjust their actions

To do better for the environment over time

Based on an objective function and feedback from the environment

Not limited only to linear systems but also other problems

as language, vision, and planning, etc.

The Foundation of AI

The Foundation of AI

Linguistics

For understanding natural languages

different approaches has been adopted from the linguistic work

Formal languages

Syntactic and semantic analysis

Knowledge representation

The main topics in AI

Artificial intelligence can be considered under a number of headings:

Search (includes Game Playing).

Representing Knowledge and Reasoning with it.

Planning.

Learning.

Natural language processing.

Expert Systems.

Interacting with the Environment
(e.g. Vision, Speech recognition, Robotics)

Search

Search is the fundamental technique of AI.

Possible answers, decisions or courses of action are structured into an abstract space, which we then search.

Search is either "blind" or “uninformed":

blind

we move through the space without worrying about what is coming next, but recognising the answer if we see it

informed

we guess what is ahead, and use that information to decide where to look next.

We may want to search for the first answer that satisfies our goal, or we may want to keep searching until we find the best answer.

Knowledge Representation & Reasoning

The second most important concept in AI

If we are going to act rationally in our environment, then we must have some way of describing that environment and drawing inferences from that representation.

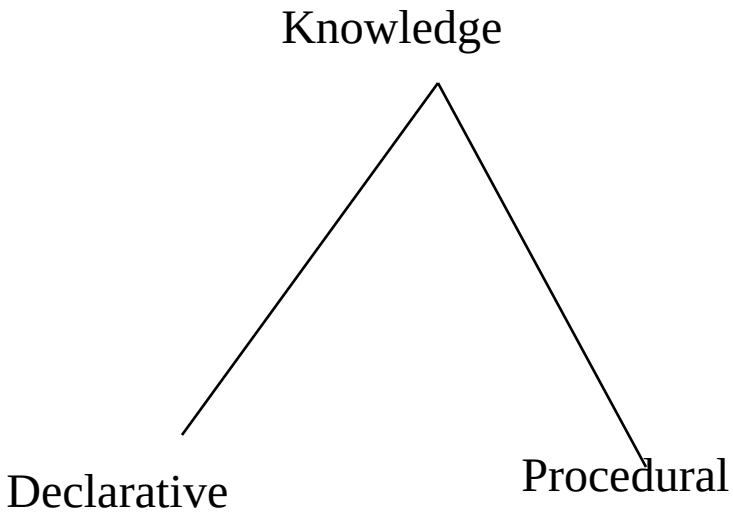
how do we describe what we know about the world ?

how do we describe it *concisely* ?

how do we describe it so that we can get hold of the right piece of knowledge when we need it ?

how do we generate new pieces of knowledge ?

how do we deal with *uncertain* knowledge ?



• Declarative knowledge deals with factoid questions (what is the capital of India? Etc.)

• Procedural knowledge deals with “How”

• Procedural knowledge can be embedded in declarative knowledge

Planning

Given a set of goals, construct a sequence of actions that achieves those goals:

often very large search space

but most parts of the world are independent of most other parts

often start with goals and connect them to actions

no necessary connection between order of planning and order of execution

what happens if the world changes as we execute the plan and/or our actions don't produce the expected results?

Learning

- If a system is going to act truly appropriately, then it must be able to change its actions in the light of experience:
  - how do we generate new facts from old ?
  - how do we generate new concepts ?
  - how do we learn to distinguish different situations in new environments ?

Interacting with the Environment

- In order to enable intelligent behaviour, we will have to interact with our environment.
- Properly intelligent systems may be expected to:
  - accept sensory input
    - vision, sound, …
  - interact with humans
    - understand language, recognise speech, generate text, speech and graphics, …
  - modify the environment
    - robotics

History of AI

AI has a long history

Ancient Greece

Aristotle

Historical Figures Contributed

Ramon Lull

Al Khowarazmi

Leonardo da Vinci

David Hume

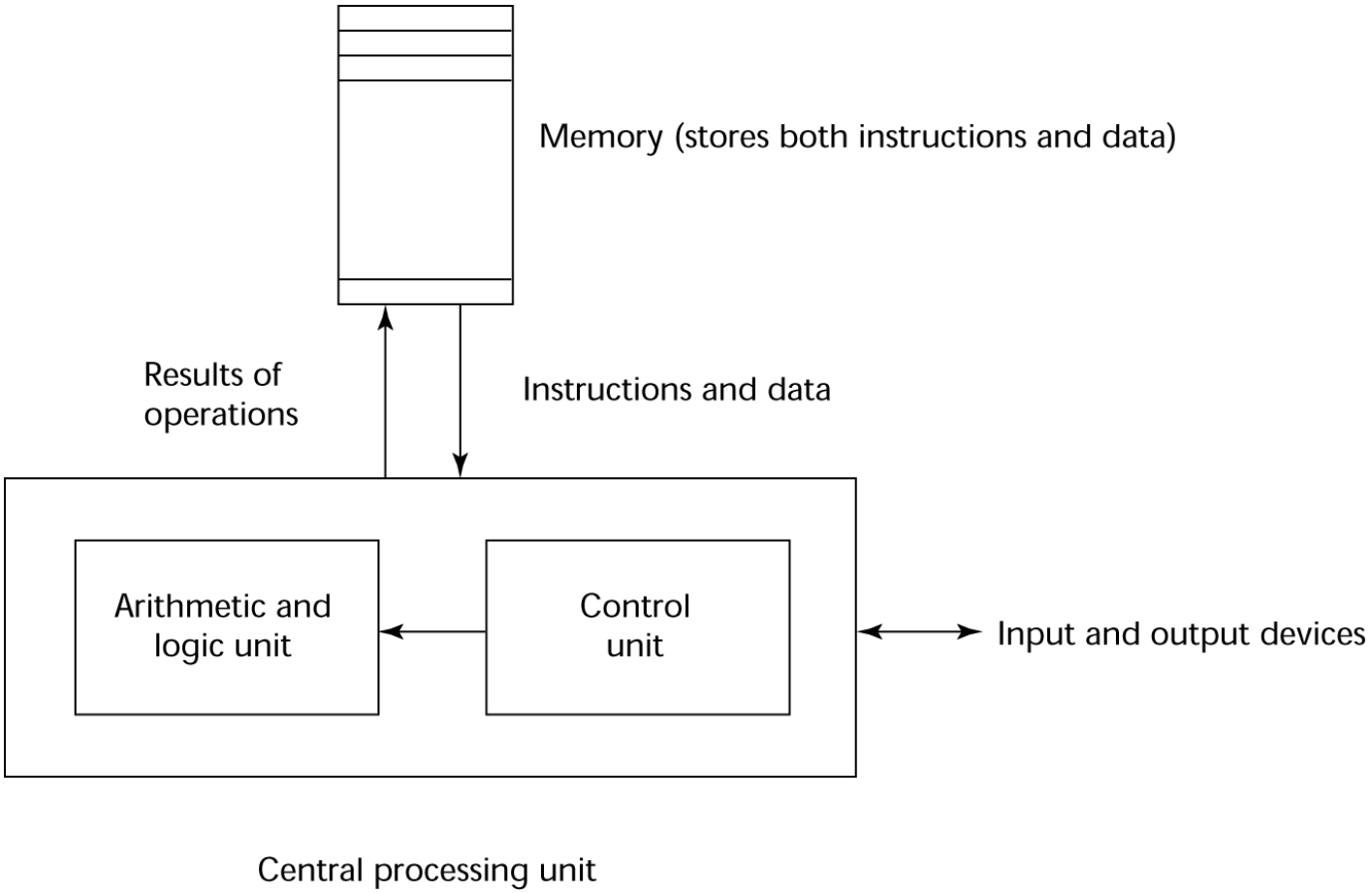
George Boole

Charles Babbage

John von Neuman

As old as electronic computers themselves (c1940)

The ‘von Neuman’ Architecture



History of AI

Origins

The Dartmouth conference: 1956

John McCarthy (Stanford)

Marvin Minsky (MIT)

Herbert Simon (CMU)

Allen Newell (CMU)

Arthur Samuel (IBM)

The Turing Test (1950)

“Machines who Think”

By Pamela McCorckindale

Periods in AI

Early period - 1950's & 60's

Game playing

brute force (calculate your way out)

Theorem proving

symbol manipulation

Biological models

neural nets

Symbolic application period - 70's

Early expert systems, use of knowledge

Commercial period - 80's

boom in knowledge/ rule bases

Periods in AI cont'd

? period - 90's and New Millenium

Real-world applications, modelling, better evidence, use of theory, ……?

Topics: data mining, formal models, GA's, fuzzy logic, agents, neural nets, autonomous systems

Applications

visual recognition of traffic

medical diagnosis

directory enquiries

power plant control

automatic cars

Symbolic and Sub-symbolic AI

Symbolic AI is concerned with describing and manipulating our knowledge of the world as explicit symbols, where these symbols have clear relationships to entities in the real world.

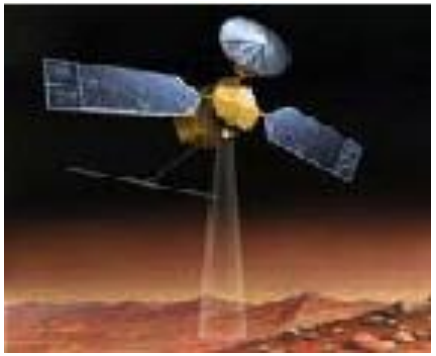
Sub-symbolic AI (e.g. neural-nets) is more concerned with obtaining the correct response to an input stimulus without 'looking inside the box' to see if parts of the mechanism can be associated with discrete real world objects.

This course is concerned with symbolic AI.

AI Applications

- Autonomous Planning & Scheduling:
  - Autonomous rovers.



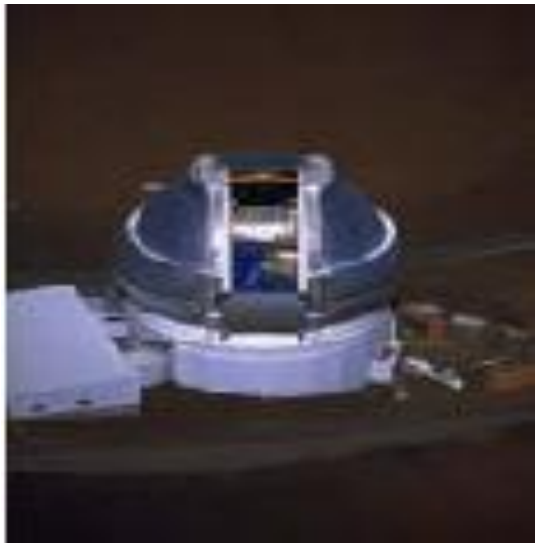


Autonomous Planning & Scheduling:

Telescope scheduling

AI Applications



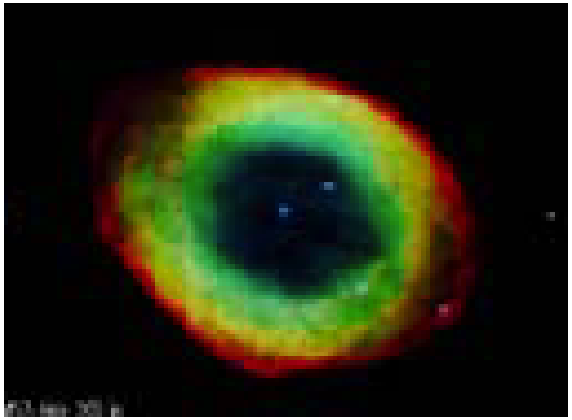


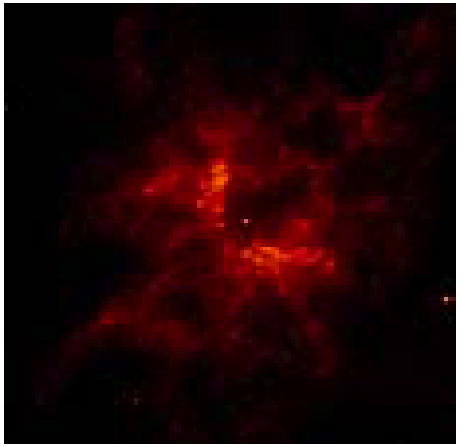
Autonomous Planning & Scheduling:

Analysis of data:

AI Applications



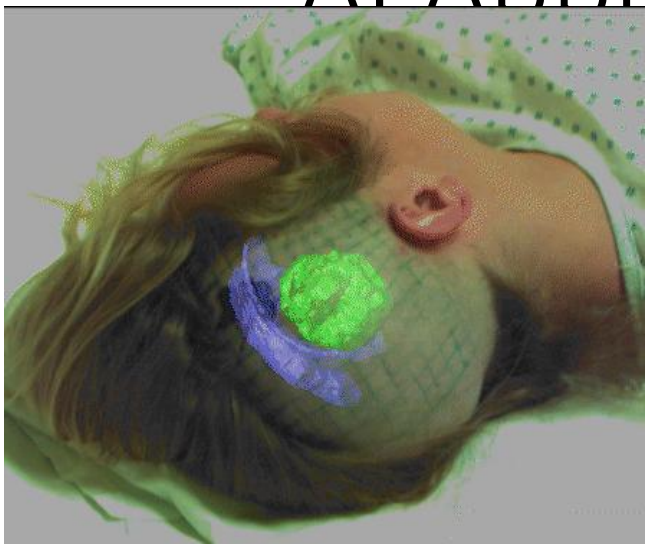


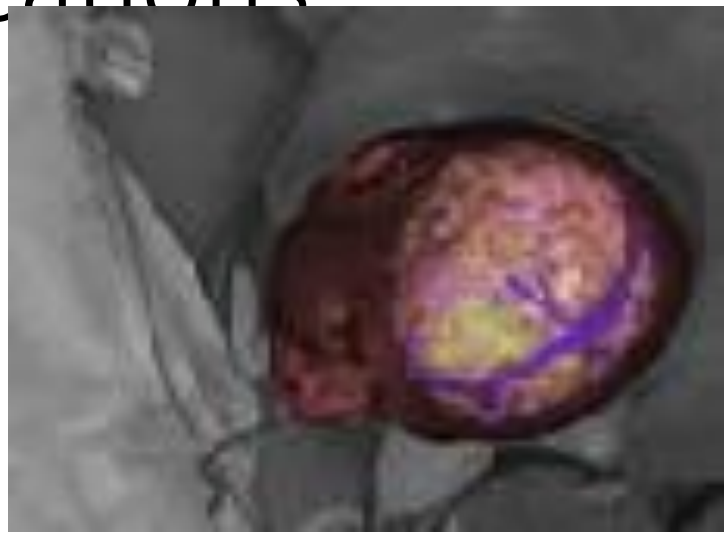


Medicine:

Image guided surgery

AI Applications

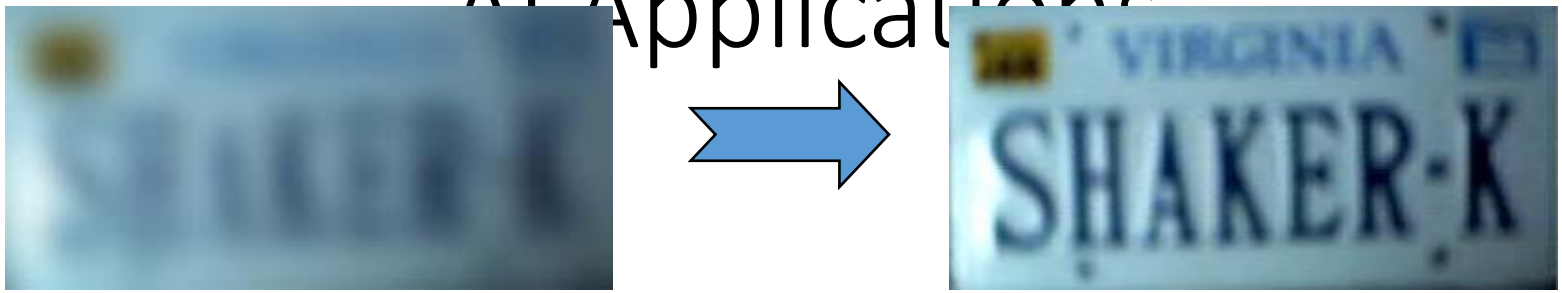


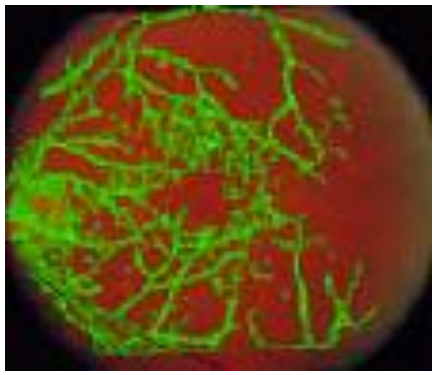


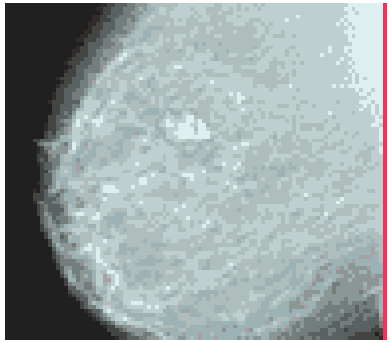
Medicine:

Image analysis and enhancement

AI Applications







AI Applications

- **Transportation**:
  - **Autonomous vehicle control:**



Transportation:

Pedestrian detection:

AI Applications

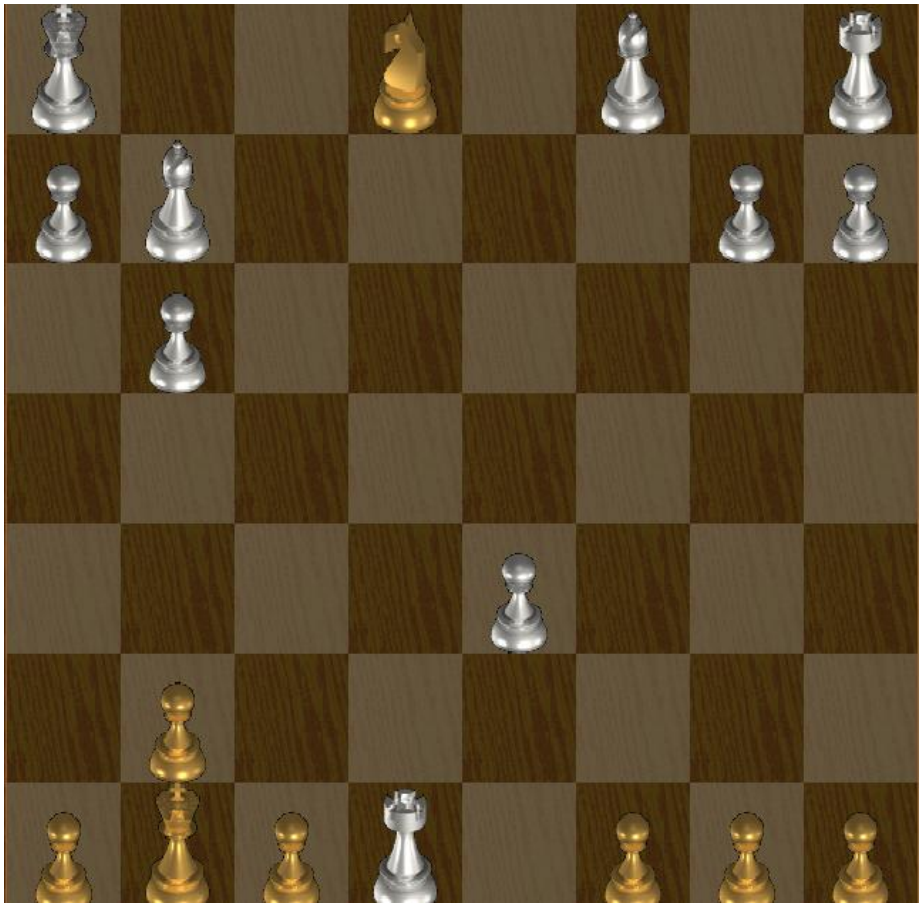


Games:



AI Applications

Games:



Robotic toys:



Applications



AI Applications

Other application areas:

Bioinformatics:

Gene expression data analysis

Prediction of protein structure

Text classification, document sorting:

Web pages, e-mails

Articles in the news

Video, image classification

Music composition, picture drawing

Natural Language Processing .

Perception.

Why Artificial Intelligence?

* Before Learning about Artificial Intelligence, we should know that what is the importance of AI and why should we learn it. Following are some main reasons to learn about AI:
* With the help of AI, you can create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
* With the help of AI, you can create your personal virtual Assistant, such as Cortana, Google Assistant, Siri, etc.

Why Artificial Intelligence?

* With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.
* AI opens a path for other new technologies, new devices, and new Opportunities.

Goals of Artificial Intelligence ?

* Replicate human intelligence.
* Solve Knowledge-intensive tasks.
* An intelligent connection of perception and action.
* Building a machine which can perform tasks that requires human intelligence such as:
  * Proving a theorem
  * Playing chess
  * Plan some surgical operation
  * Driving a car in traffic
* Creating some system which can exhibit intelligent behavior, learn new things by itself, demonstrate, explain, and can advise to its user.

What Comprises to Artificial Intelligence?

* Artificial Intelligence is not just a part of computer science even it's so vast and requires lots of other factors which can contribute to it.
* To create the AI first we should know that how intelligence is composed, so the Intelligence is an intangible part of our brain which is a combination of **Reasoning, learning, problem-solving perception, language understanding, etc**.

What Comprises to Artificial Intelligence?

* To achieve the above factors for a machine or software Artificial Intelligence requires the following discipline:
  * Mathematics
  * Biology
  * Psychology
  * Sociology
  * Computer Science
  * Neurons Study
  * Statistics

Advantages of Artificial Intelligence

* **High Accuracy with less errors:** AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.
* **High-Speed:** AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
* **High reliability:** AI machines are highly reliable and can perform the same action multiple times with high accuracy.
* **Useful for risky areas:** AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.

Advantages of Artificial Intelligence

* **Digital Assistant:** AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.
* **Useful as a public utility:** AI can be very useful for public utilities such as a self-driving car which can make our journey safer and hassle-free, facial recognition for security purpose, Natural language processing to communicate with the human in human-language, etc.

Disadvantages of Artificial Intelligence

* **High Cost:** The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
* **Can't think out of the box:** Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.
* **No feelings and emotions:** AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.

Disadvantages of Artificial Intelligence

- **Increase dependency on machines:** With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.
- **No Original Creativity:** As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

Application of AI

* **AI in Astronomy:** Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.
* **AI in Healthcare:** In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.

  Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

Application of AI

* **AI in Gaming:** AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.
* **AI in Finance:** AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

Application of AI

* **AI in Data Security:** The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

* **AI in Social Media:** Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

Application of AI

* **AI in Travel & Transport:** AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.
* **AI in Automotive Industry:** Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.
* Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

Application of AI

* **AI in Robotics:** Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
* Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.
* **AI in Entertainment:** We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

Application of AI

* **AI in Agriculture:** Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.
* **AI in E-commerce:** AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

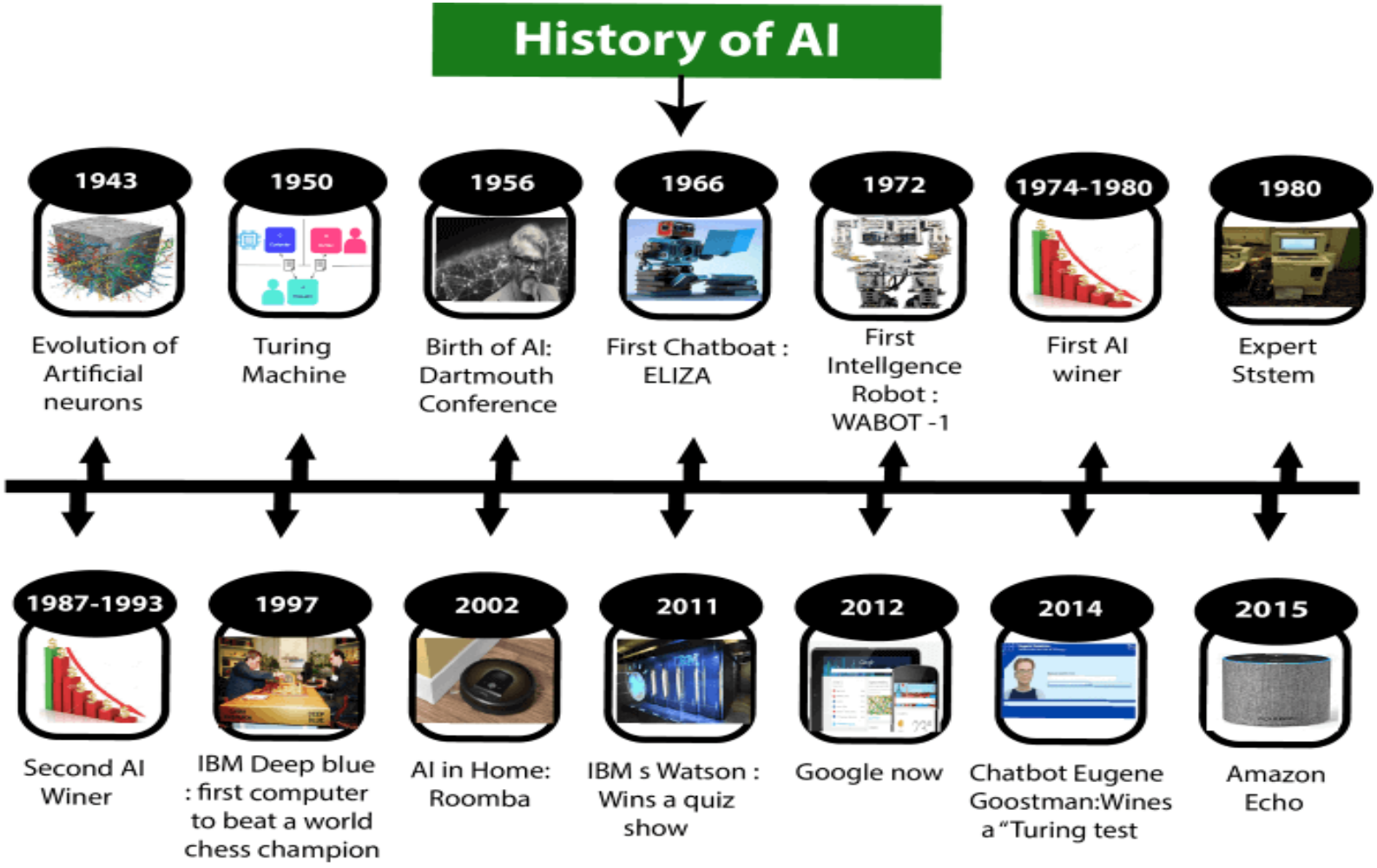
Application of AI

* **AI in education:** AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.
* AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

History of Artificial Intelligence

* Artificial Intelligence is not a new word and not a new technology for researchers.
* This technology is much older than you would imagine.
* Even there are the myths of Mechanical men in Ancient Greek and Egyptian Myths.
* Following are some milestones in the history of AI which defines the journey from the AI generation to till date development.

History of Artificial Intelligence



Maturation of Artificial Intelligence (1943-1952)

* **Year 1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter pits in 1943. They proposed a model of **artificial neurons**.

* **Year 1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called **Hebbian learning**.

* **Year 1950:** The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes **"Computing Machinery and Intelligence"** in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a **Turing test**.

The birth of Artificial Intelligence (1952-1956)

* **Year 1955:** An Allen Newell and Herbert A. Simon created the "first artificial intelligence program "Which was named as **"Logic Theorist"**. This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.

* **Year 1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first time, AI coined as an academic field**.**

* At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

The golden years-Early enthusiasm (1956-1974)

* **Year 1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.

* **Year 1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

The first AI winter (1974-1980)

* The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.

* During AI winters, an interest of publicity on artificial intelligence was decreased.

A boom of AI (1980-1987)

* **Year 1980:** After AI winter duration, AI came back with "Expert System". Expert systems were programmed that emulate the decision-making ability of a human expert.

* In the Year 1980, the first national conference of the American Association of Artificial Intelligence **was held at Stanford University**.

The second AI winter (1987-1993)

* The duration between the years 1987 to 1993 was the second AI Winter duration.

* Again Investors and government stopped in funding for AI research as due to high cost but not efficient result. The expert system such as XCON was very cost effective.

The emergence of intelligent agents (1993-2011)

* **Year 1997:** In the year 1997, IBM Deep Blue beats world chess champion, Gary Kasparov, and became the first computer to beat a world chess champion.

* **Year 2002:** for the first time, AI entered the home in the form of Roomba, a vacuum cleaner.

* **Year 2006:** AI came in the Business world till the year 2006. Companies like Facebook, Twitter, and Netflix also started using AI.

Deep learning, big data and artificial general intelligence (2011-present)

* **Year 2011:** In the year 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.

* **Year 2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.

* **Year 2014:** In the year 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test.“

Deep learning, big data and artificial general intelligence (2011-present)

* **Year 2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.

* Google has demonstrated an AI program "Duplex" which was a virtual assistant and which had taken hairdresser appointment on call, and lady on other side didn't notice that she was talking with the machine.

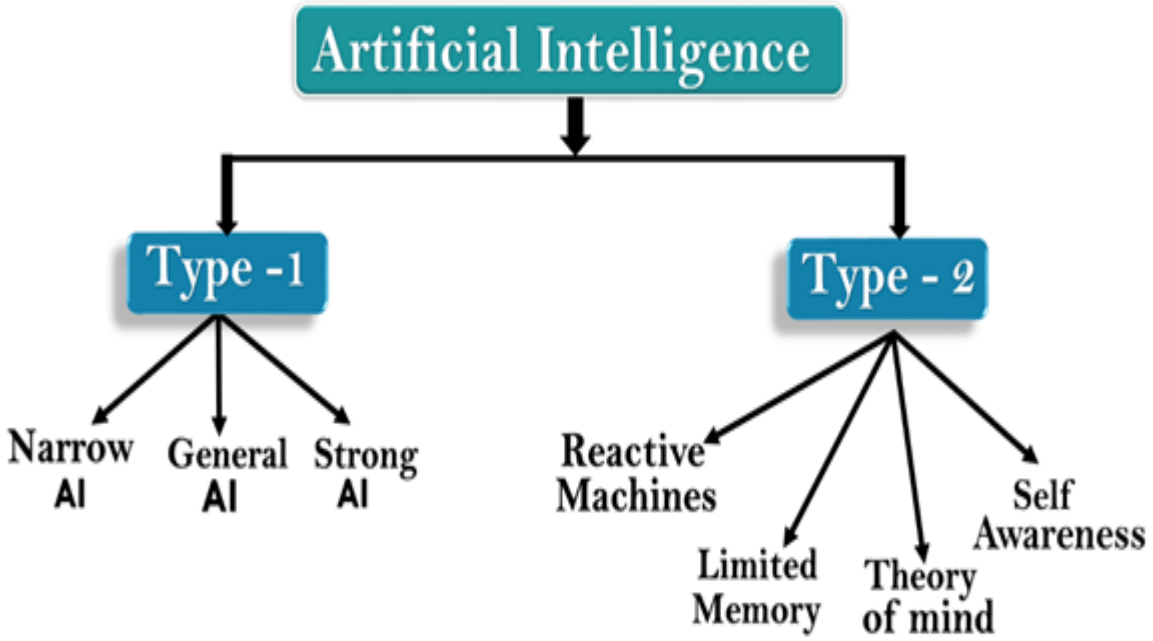
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Types of Artificial Intelligence:

* Artificial Intelligence can be divided in various types, there are mainly two types which are based on capabilities and based on functionally of AI.



AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:

* Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.
* Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.

AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:

* Apple Siri is a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
* IBM's Watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
* Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

AI type-1: Based on Capabilities

2. General AI:

* General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
* The idea behind the general AI to make such a system which could be smarter and think like a human by its own.
* Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human.
* The worldwide researchers are now focused on developing machines with General AI.
* As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.

AI type-1: Based on Capabilities

3. Super AI:

* Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an outcome of general AI.
* Some key characteristics of strong AI include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.
* Super AI is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is still world changing task.

Artificial Intelligence type-2: Based on functionality

1. Reactive Machines:

* Purely reactive machines are the most basic types of Artificial Intelligence.
* Such AI systems do not store memories or past experiences for future actions.
* These machines only focus on current scenarios and react on it as per possible best action.
* IBM's Deep Blue system is an example of reactive machines.
* Google's AlphaGo is also an example of reactive machines.

Artificial Intelligence type-2: Based on functionality

2. Limited Memory:

* Limited memory machines can store past experiences or some data for a short period of time.
* These machines can use stored data for a limited time period only.
* Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

Artificial Intelligence type-2: Based on functionality

3. Theory of Mind:

* Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.

* This type of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

Artificial Intelligence type-2: Based on functionality

4. Self-Awareness:

* Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.

* These machines will be smarter than human mind.

* Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

Types of AI Agents

* Agents can be grouped into five classes based on their degree of perceived intelligence and capability.
* All these agents can improve their performance and generate better action over the time. These are given below:
  * Simple Reflex Agent
  * Model-based reflex agent
  * Goal-based agents
  * Utility-based agent
  * Learning agent

Types of AI Agents

1. Simple Reflex agent:

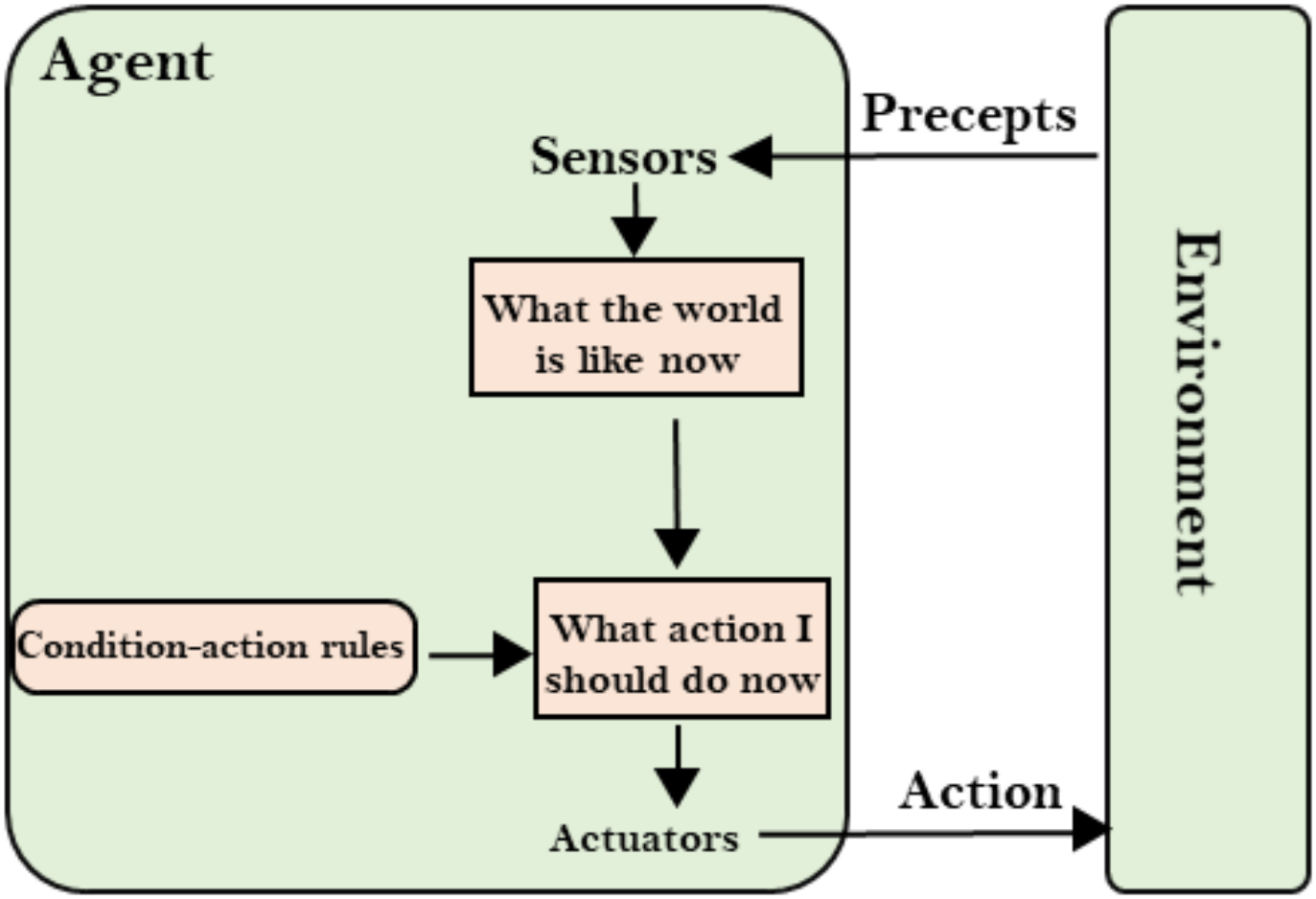
* The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history.
* These agents only succeed in the fully observable environment.
* The Simple reflex agent does not consider any part of percepts history during their decision and action process.
* The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.

Types of AI Agents

1. Simple Reflex agent:

* Problems for the simple reflex agent design approach.
  * They have very limited intelligence
  * They do not have knowledge of non-perceptual parts of the current state
  * Mostly too big to generate and to store.
  * Not adaptive to changes in the environment.

Types of AI Agents

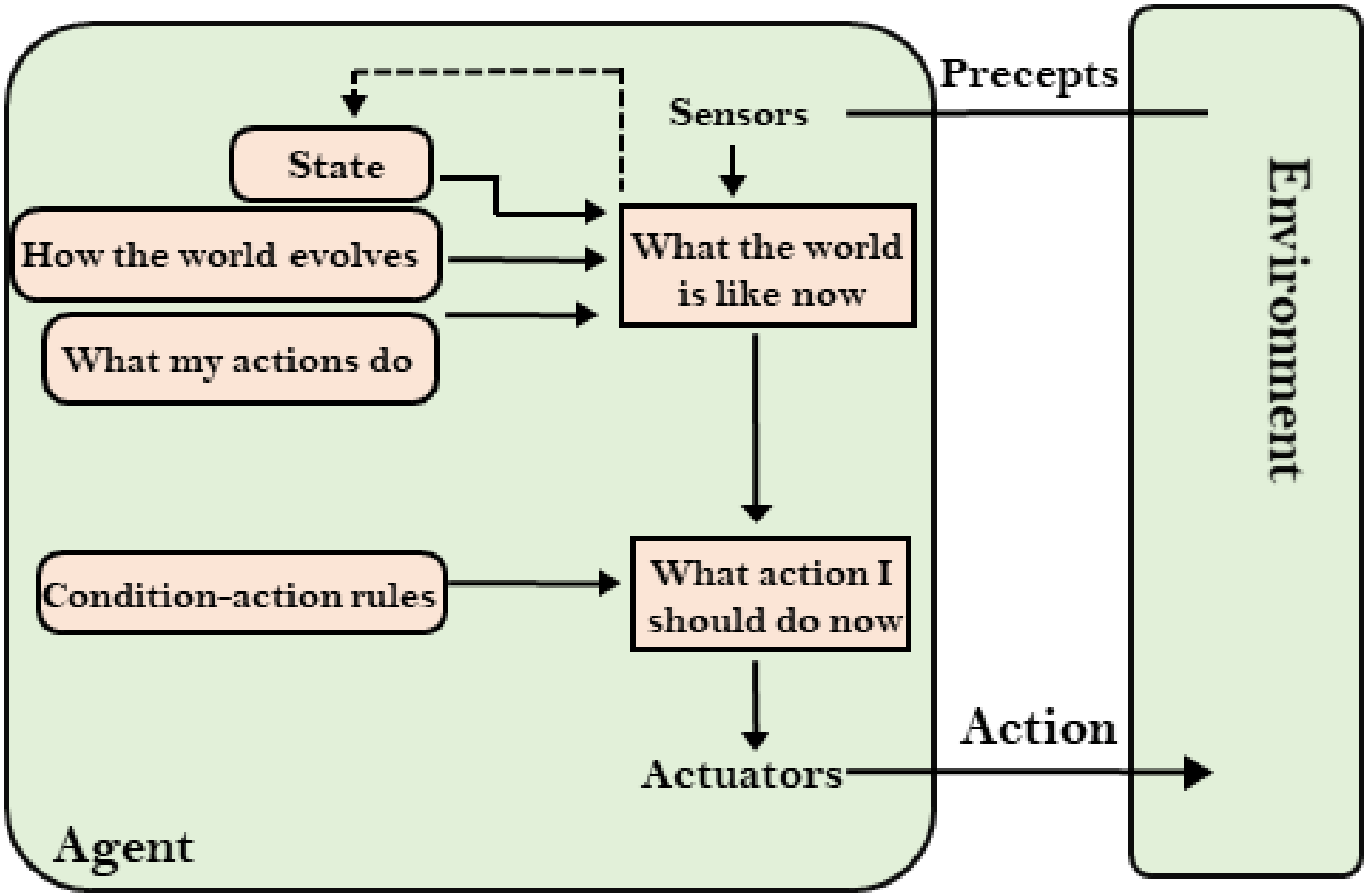


Types of AI Agents

2. Model-based reflex agent:

* The Model-based agent can work in a partially observable environment, and track the situation.
* A model-based agent has two important factors:
  * **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  * **Internal State:** It is a representation of the current state based on percept history.
* These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
* Updating the agent state requires information about:
  * How the world evolves
  * How the agent's action affects the world.

Types of AI Agents

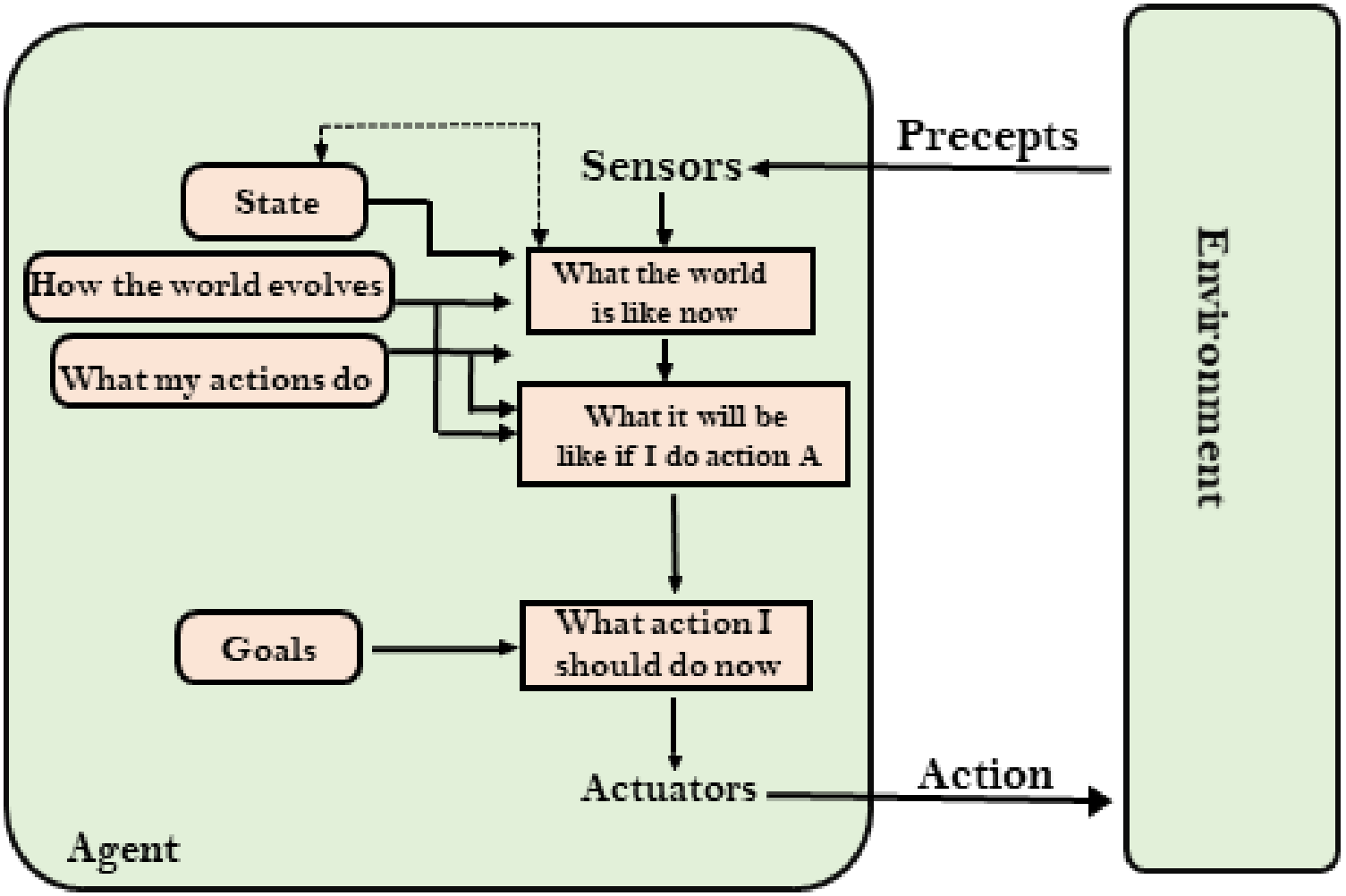


Types of AI Agents

3. Goal-based agents:

* The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
* The agent needs to know its goal which describes desirable situations.
* Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
* They choose an action, so that they can achieve the goal.
* **These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.**

Types of AI Agents

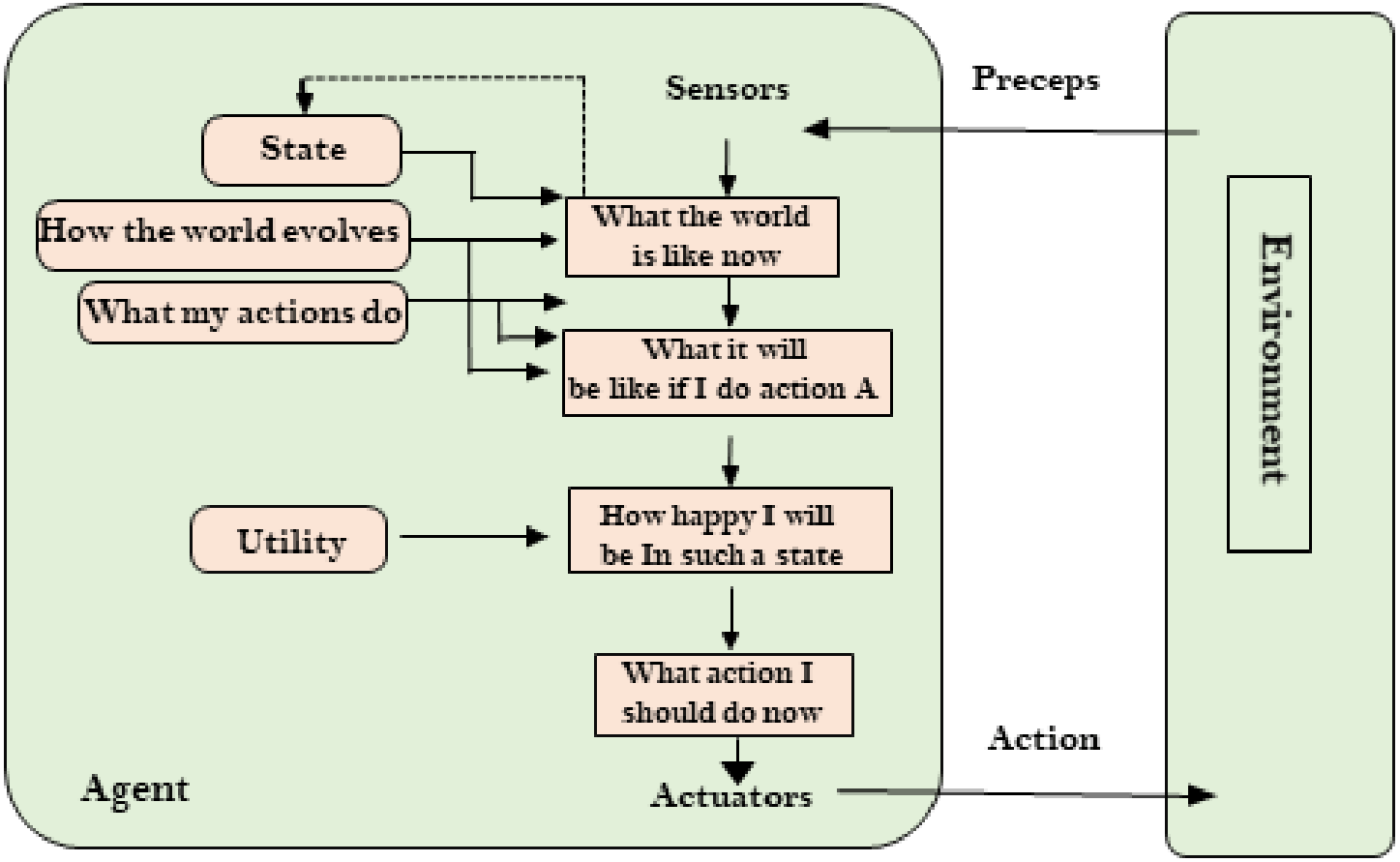


Types of AI Agents

4. Utility-based agents

* These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
* Utility-based agent acts based not only on goals but also the best way to achieve the goal.
* The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
* **The utility function maps each state to a real number to check how efficiently each action achieves the goals.**

Types of AI Agents



Types of AI Agents

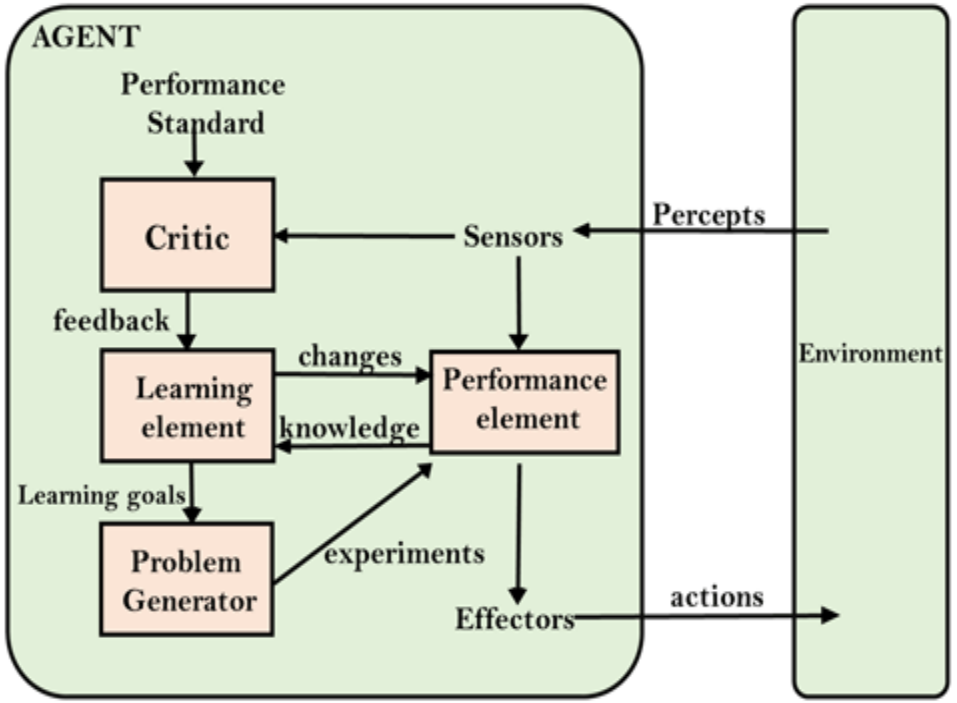
5. Learning Agents

* A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
* It starts to act with basic knowledge and then able to act and adapt automatically through learning.
* A learning agent has mainly four conceptual components, which are:
  * **Learning element:** It is responsible for making improvements by learning from environment
  * **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  * **Performance element:** It is responsible for selecting external action
  * **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Types of AI Agents

5. Learning Agents

* Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.



Agents in Artificial Intelligence

* An AI system can be defined as the study of the rational agent and its environment.
* The agents sense the environment through sensors and act on their environment through actuators.
* An AI agent can have mental properties such as knowledge, belief, intention, etc.

What is an Agent?

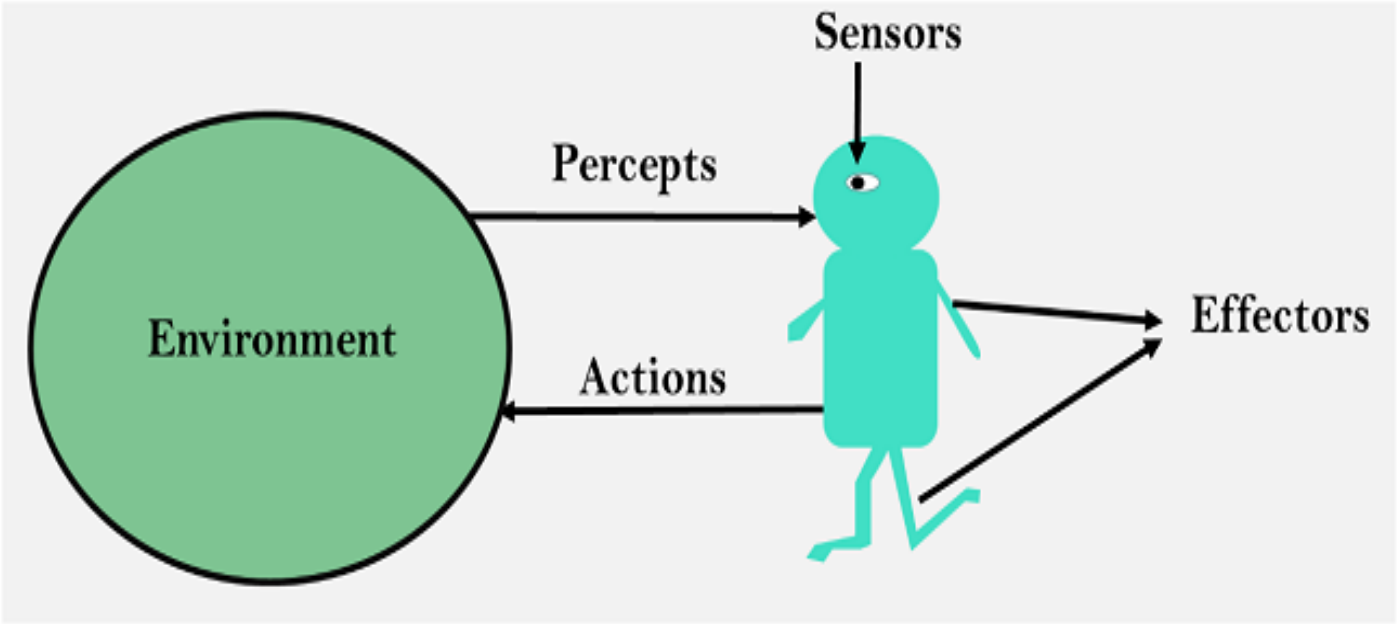
* An agent can be anything that perceive its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of **perceiving**, **thinking**, and **acting**. An agent can be:
  * **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
  * **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
  * **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

What is an Agent?

* Hence the world around us is full of agents such as thermostat, cellphone, camera, and even we are also agents.
* Before moving forward, we should first know about sensors, effectors, and actuators.
* **Sensor:** Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.
* **Actuators:** Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

What is an Agent?

* **Effectors:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.



Intelligent Agents:

* An intelligent agent is an autonomous entity which act upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.
* Following are the main four rules for an AI agent:
  * **Rule 1:** An AI agent must have the ability to perceive the environment.
  * **Rule 2:** The observation must be used to make decisions.
  * **Rule 3:** Decision should result in an action.
  * **Rule 4:** The action taken by an AI agent must be a rational action.

Rational Agent:

* A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.
* A rational agent is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.
* For an AI agent, the rational action is most important because in AI reinforcement learning algorithm, for each best possible action, agent gets the positive reward and for each wrong action, an agent gets a negative reward.

Note: Rational agents in AI are very similar to intelligent agents.

Rationality:

* The rationality of an agent is measured by its performance measure.
* Rationality can be judged on the basis of following points:
  * Performance measure which defines the success criterion.
  * Agent prior knowledge of its environment.
  * Best possible actions that an agent can perform.
  * The sequence of percepts.

Note: Rationality differs from Omniscience because an Omniscient agent knows the actual outcome of its action and act accordingly, which is not possible in reality.

Structure of an AI Agent:

* The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

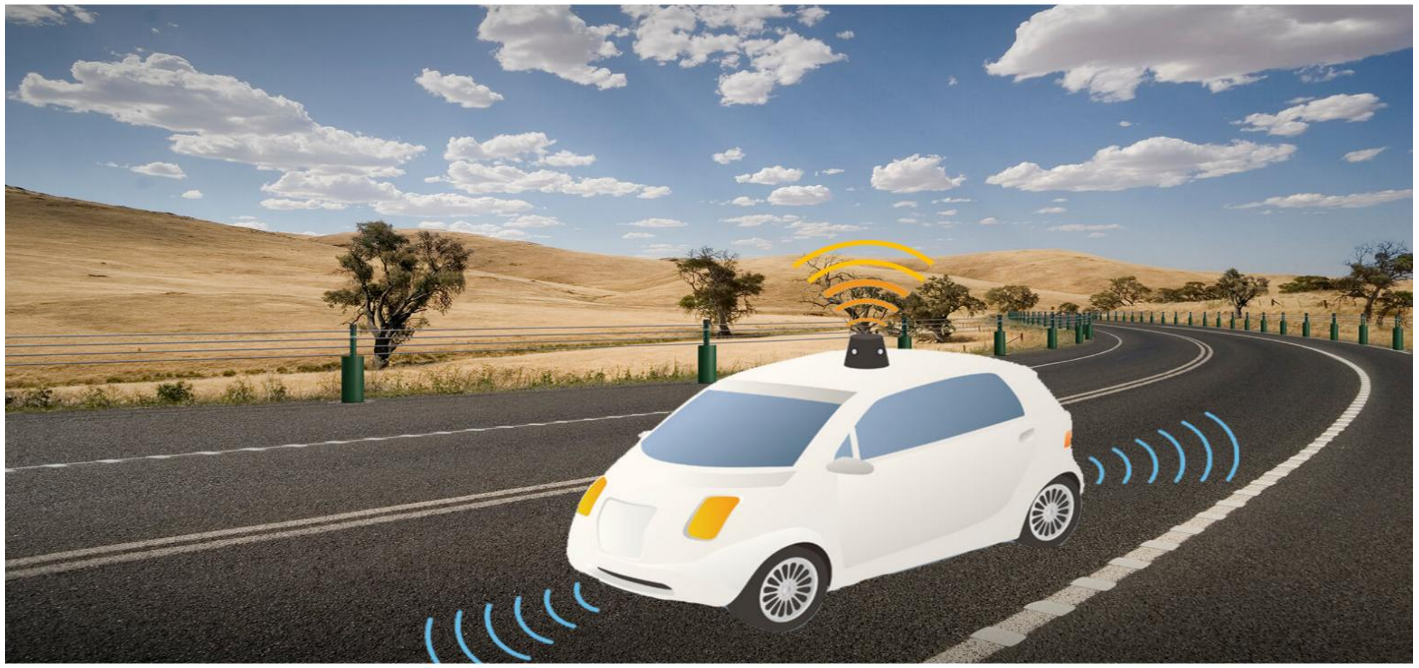
  Agent = Architecture + Agent program

* Following are the main three terms involved in the structure of an AI agent:
  * **Architecture:** Architecture is machinery that an AI agent executes on.
  * **Agent Function:** Agent function is used to map a percept to an action.
  * **Agent program:** Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f.

PEAS Representation

* PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:
  * **P:** Performance measure
  * **E:** Environment
  * **A:** Actuators
  * **S:** Sensors
* Here performance measure is the objective for the success of an agent's behavior.

PEAS for self-driving cars:



PEAS for self-driving cars:

* Let's suppose a self-driving car then PEAS representation will be:
* **Performance:** Safety, time, legal drive, comfort
* **Environment:** Roads, other vehicles, road signs, pedestrian
* **Actuators:** Steering, accelerator, brake, signal, horn
* **Sensors:** Camera, GPS, speedometer, odometer, accelerometer, sonar.

Agent Environment in AI:

* An environment is everything in the world which surrounds the agent, but it is not a part of an agent itself.
* An environment can be described as a situation in which an agent is present.
* The environment is where agent lives, operate and provide the agent with something to sense and act upon it.
* An environment is mostly said to be non-feministic.

Features of Environment

* As per Russell and Norvig, an environment can have various features from the point of view of an agent:
  * Fully observable vs Partially Observable
  * Static vs Dynamic
  * Discrete vs Continuous
  * Deterministic vs Stochastic
  * Single-agent vs Multi-agent
  * Episodic vs sequential
  * Known vs Unknown
  * Accessible vs Inaccessible

1. Fully observable vs Partially Observable:

* If an agent sensor can sense or access the complete state of an environment at each point of time then it is **a fully observable** environment, else it is **partially observable**.
* A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
* An agent with no sensors in all environments then such an environment is called as **unobservable**.

2. Deterministic vs Stochastic:

* If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.
* A stochastic environment is random in nature and cannot be determined completely by an agent.
* In a deterministic, fully observable environment, agent does not need to worry about uncertainty.

3. Episodic vs Sequential:

* In an episodic environment, there is a series of one-shot actions, and only the current percept is required for the action. . **(E.g., classifying images.)**

* However, in Sequential environment, an agent requires memory of past actions to determine the next best actions. **(E.g., chess and driving)**

4. Single-agent vs Multi-agent:

* If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.

* However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.

* The agent design problems in the multi-agent environment are different from single agent environment.

5. Static vs Dynamic:

* If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
* Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
* However for dynamic environment, agents need to keep looking at the world at each action.
* **Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.**
    - **The environment is semidynamic if the environment itself does not change with the passage of time but the agent's performance score does.**

6. Discrete vs Continuous:

* If in an environment there are a finite number of percepts and actions that can be performed within it, then such an environment is called a discrete environment else it is called continuous environment.
* **A chess game comes under discrete environment as there is a finite number of moves that can be performed.**
* **A self-driving car is an example of a continuous environment.**

7. Known vs Unknown:

* Known and unknown are not actually a feature of an environment, but it is an agent's state of knowledge to perform an action.
* In a known environment, the results for all actions are known to the agent. While in unknown environment, agent needs to learn how it works in order to perform an action.
* It is quite possible that a known environment to be partially observable and an Unknown environment to be fully observable.

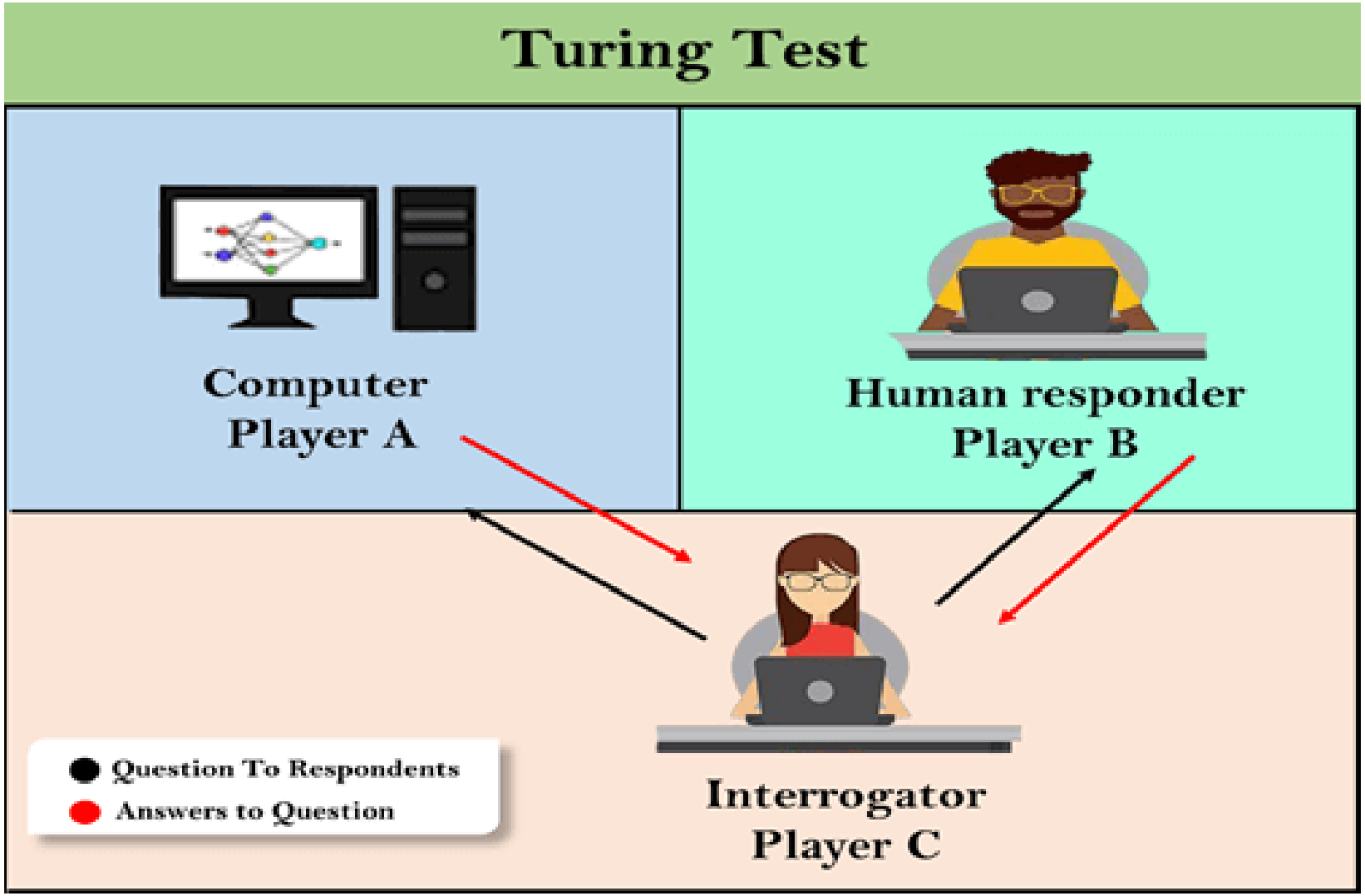
Accessible vs Inaccessible:

* If an agent can obtain complete and accurate information about the state's environment, then such an environment is called an Accessible environment else it is called inaccessible.
* **An empty room whose state can be defined by its temperature is an example of an accessible environment.**
* **Information about an event on earth is an example of Inaccessible environment.**

Turing Test in AI:

* In 1950, Alan Turing introduced a test to check whether a machine can think like a human or not, this test is known as the Turing Test.
* In this test, Turing proposed that the computer can be said to be an intelligent if it can mimic human response under specific conditions.
* Turing Test was introduced by Turing in his 1950 paper, "Computing Machinery and Intelligence," which considered the question, "Can Machine think?"

Turing Test in AI:



Turing Test in AI:

* The Turing test is based on a party game "Imitation game," with some modifications.
* This game involves three players in which one player is Computer, another player is human responder, and the third player is a human Interrogator, who is isolated from other two players and his job is to find that which player is machine among two of them.
* Consider, Player A is a computer, Player B is human, and Player C is an interrogator.
* Interrogator is aware that one of them is machine, but he needs to identify this on the basis of questions and their responses.

Turing Test in AI:

* The conversation between all players is via keyboard and screen so the result would not depend on the machine's ability to convert words as speech.
* The test result does not depend on each correct answer, but only how closely its responses like a human answer.
* The computer is permitted to do everything possible to force a wrong identification by the interrogator.
* The questions and answers can be like

Turing Test in AI:

* **Interrogator:** Are you a computer?
* **PlayerA (Computer):** No
* **Interrogator:** Multiply two large numbers such as (256896489*456725896)
* **Player A:** Long pause and give the wrong answer.
* In this game, if an interrogator would not be able to identify which is a machine and which is human, then the computer passes the test successfully, and the machine is said to be intelligent and can think like a human.
* "In 1991, the New York businessman Hugh Loebner announces the prize competition, offering a $100,000 prize for the first computer to pass the Turing test. However, no AI program to till date, come close to passing an undiluted Turing test".

Chatbots to attempt the Turing test:

* **ELIZA:** ELIZA was a Natural language processing computer program created by Joseph Weizenbaum. It was created to demonstrate the ability of communication between machine and humans. It was one of the first chatterbots, which has attempted the Turing Test.

* **Parry:** Parry was a chatterbot created by Kenneth Colby in 1972. Parry was designed to simulate a person with **Paranoid schizophrenia**(most common chronic mental disorder). Parry was described as "ELIZA with attitude." Parry was tested using a variation of the Turing Test in the early 1970s.

Chatbots to attempt the Turing test::

* **Eugene Goostman:** Eugene Goostman was a chatbot developed in Saint Petersburg in 2001.
* This bot has competed in the various number of Turing Test.
* In June 2012, at an event, Goostman won the competition promoted as largest-ever Turing test content, in which it has convinced 29% of judges that it was a human.
* Goostman resembled as a 13-year old virtual boy.

The Chinese Room Argument:

* There were many philosophers who really disagreed with the complete concept of Artificial Intelligence. The most famous argument in this list was "**Chinese Room**.“
* In the year **1980, John Searle** presented "**Chinese Room**" thought experiment, in his paper "**Mind, Brains, and Program**," which was against the validity of Turing's Test.
* According to his argument, "**Programming a computer may make it to understand a language, but it will not produce a real understanding of language or consciousness in a computer**."

The Chinese Room Argument:

* He argued that Machine such as ELIZA and Parry could easily pass the Turing test by manipulating keywords and symbol, but they had no real understanding of language.
* So it cannot be described as "thinking" capability of a machine such as a human.

Features required for a machine to pass the Turing test:

- **Natural language processing:** NLP is required to communicate with Interrogator in general human language like English.
- **Knowledge representation:** To store and retrieve information during the test.
- **Automated reasoning:** To use the previously stored information for answering the questions.
- **Machine learning:** To adapt new changes and can detect generalized patterns.
- **Vision (For total Turing test):** To recognize the interrogator actions and other objects during a test.
- **Motor Control (For total Turing test):** To act upon objects if requested.

THANK YOU